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Lecture 1: Optimal Transport

Lecture 2: Generative Flows

Lecture 3: From 1D Processes to Flows

Lecture 4: Bayesian Inverse Problems

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GENERATIVE FLOWS FROM 1D PROCESSES

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Joint work with

Jannis Chemseddine, Richard Duong, Gregor Kornhardt and Peter Friz



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Task: similar as in diffusion models

- ◆ Learn transport of samples from $\mu_0 = \mu_X$ to samples of $\mu_1 = \mu_Z$ (target to noise)
Learn/approximate v_t by neural network v_t^θ !
- ◆ Inference: use Backward flow with $-v_{1-t}$ in ODE (noise to target)

Loss function for learning:

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0, T), x_0 \sim \mu_0, x \sim \mu_t(\cdot | x_0)} \left[\|v_t^\theta(x) - v_t^{x_0}\|^2 \right]$$



All you need is conditional $v_t^{x_0}$!

Recall:

- ◆ We have already seen how to get $v_t^{x_0}$ from couplings, resp. linear stochastic processes.
- ◆ So far our latent distribution was a standard Gaussian.

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Outline

1. From One-Dimensional to Multi-Dimensional Flows
2. 1D Flows Besides Diffusion
3. Conclusions

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1. **One-dimensional noise**: construct flows in 1D fulfilling

$$\partial_t \mu_t + \partial_x(\mu_t v_t) = 0, \quad \mu_0 = \delta_0 \quad (\text{CE})$$

2. **Multi-dimensional noise**: set up a multi-dimensional flow starting in

$$\mu_0 = \delta_0, \quad 0 \in \mathbb{R}^d$$

3. **Incorporating the data**: find a multi-dimensional flow starting in

$$\mu_0 = \delta_{x_0}, \quad x_0 \sim \mu_0$$

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Step 2: Multi-Dimensional Noise

Motivation: Wiener process $\mathbf{W}_t = (W_t^1, \dots, W_t^d) \in \mathbb{R}^d$ consists of independent, identically distributed 1D components

Proposition

For $i = 1, \dots, d$, let (μ_t^i, v_t^i) fulfill the **one-dimensional CE** and the μ_t^i are mutually independent. Then

$$\mu_t(x) = \prod_{i=1}^d \mu_t^i(x^i)$$

has process $\mathbf{Y}_t := (Y_t^1, \dots, Y_t^d) \sim \mu_t$ and satisfies a **multi-dimensional CE** with a velocity field

$$v_t(x) := (v_t^1(x^1), \dots, v_t^d(x^d)) .$$

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Step 3: Incorporating the Data

Done: $(\mathbf{Y}_t)_t$ starting in $\mathbf{Y}_0 = 0 \in \mathbb{R}^d$ with $v_t = v_t^{\mathbf{Y}}(\cdot \mid 0)$

Task: $(\mathbf{X}_t)_t$ which can start in any sample x_0 from μ_0 . Let $\mathbf{X}_0 \sim \mu_0$.

Mean-reverting process: $\mathbf{X}_t := f(t) \mathbf{X}_0 + \mathbf{Y}_{g(t)}, t \in [0, 1]$

with smooth **scheduling functions** f, g satisfying

$$f(0) = 1, \quad f(1) = 0 \quad \text{and} \quad g(0) = 0, \quad g(1) = 1$$

Conditional velocity field of $\mathbf{X}_t$:

$$\begin{aligned} v_t^{\mathbf{X}}(x \mid x_0) &= \mathbb{E}[\dot{\mathbf{X}}_t \mid \mathbf{X}_t = x, \mathbf{X}_0 = x_0] \\ &= \mathbb{E}[\dot{f}(t) x_0 + \dot{g}(t) \dot{\mathbf{Y}}_{g(t)} \mid \mathbf{Y}_{g(t)} = x - f(t)x_0] \\ &= \dot{f}(t) x_0 + \dot{g}(t) v_{g(t)}^{\mathbf{Y}}(x - f(t)x_0 \mid 0). \end{aligned}$$

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Step 3: Incorporating the Data

Remark (Generalization of FM and Diffusion):

$$\mathbf{X}_t = f(t) \mathbf{X}_0 + \mathbf{Y}_{g(t)}$$
$$\mathbf{X}_t^{\text{GF}} = \alpha_t \mathbf{X}_0 + \sigma_t \mathbf{X}_1, \quad \mathbf{X}_1 \sim \mathcal{N}(0, I_d)$$

◆ $f(t) := \alpha_t$, $g(t) := \sigma_t^2$, $\mathbf{Y}_t$ standard Brownian motion: $\mathbf{X}_t^{\text{GF}} \stackrel{d}{=} \mathbf{X}_t$

◆ **FM**: $f(t) := 1 - t$, $g(t) := t^2$

◆ **Diffusion (Ohrenstein-Uhlenbeck)**:

$$f(t) := \exp\left(-\frac{h(t)}{2}\right), \quad g(t) := 1 - \exp(-h(t)),$$

$$\text{where } h(t) := \int_0^t \beta_{\min} + s(\beta_{\max} - \beta_{\min}) \, ds$$

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1. From One-Dimensional to Multi-Dimensional Flows

2. 1D Flows Besides Diffusion

2.1 Prescribed Processes

2.2 Learned Processes

3. Conclusions

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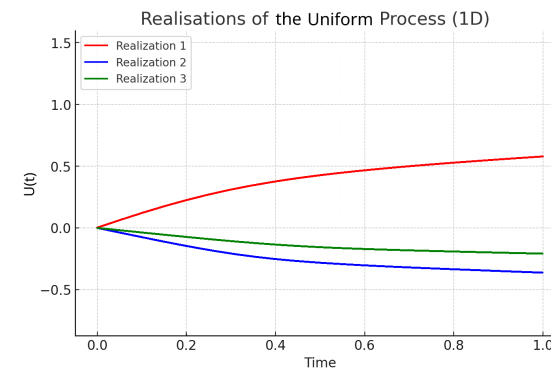
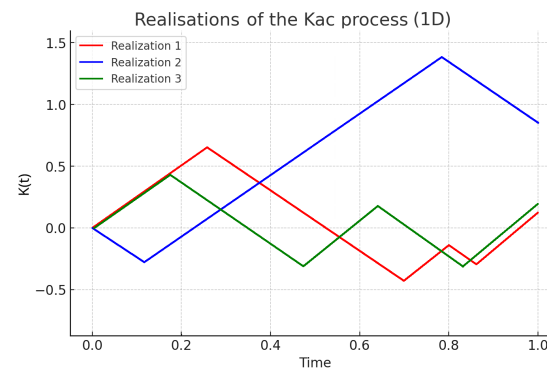
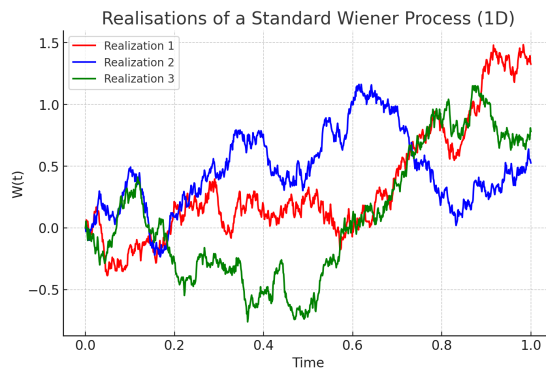
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2.1. Prescribed Processes

1. Wiener process W_t – diffusion equation
2. Kac process K_t – damped wave equation
3. Uniform process U_t – gradient flow of the maximum mean discrepancy functional $\mathcal{F}_\nu = \text{MMD}_K(\cdot, \nu)$ with negative distance kernel $K(x, y) = -|x - y|$ and $\nu = \mathcal{U}(-b, b)$, i.e. (μ_t, v_t) fulfill the continuity equation with $v_t \in -\partial\mathcal{F}_\nu(\mu_t)$



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1. Wiener Process and Diffusion Equation

Probability density flow p_t :

$$\partial_t p_t = \nabla \cdot \underbrace{\left(p_t \frac{1}{2} \nabla \log p_t \right)}_{v_t} = \frac{1}{2} \Delta p_t, \quad t \in (0, 1], \quad \lim_{t \downarrow 0} p_t = \delta_0,$$

Analytic solution:

$$p_t(x) = (2\pi t)^{-\frac{d}{2}} e^{-\frac{\|x\|^2}{2t}}, \quad p_1 = \mathcal{N}(0, I_d)$$

Velocity field:

$$v_t(x) = -\frac{1}{2} \nabla \log p_t = \frac{x}{2t}$$

Reverse velocity field:

$$-v_{1-t}(x) = -\frac{x}{2(1-t)} \longrightarrow \|v_{1-t}\|_{L_2(\mu_t)}^2 = \frac{d}{4(1-t)} \notin L_2(0, 1) \text{ 😞}$$

Instability issues caused by this explosion at times close to the target Kim et al. ICLR 2022 and similar with drifts (Fokker-Planck equation) Pidstrigach, NeuRips 2022; may be avoided by e.g. time truncations

Refs: Mate/Fleuret, TMLR 2023, Maurais/Marzuok ICML 2024, Chemsedine, Wald, Duong, St. ICLR 2025, Berner, Richter et al. 2024, ICLR 2025)

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2. Kac Process and Damped Wave Equation

Motivation

- ◆ It appears to be **unphysical** that particles in diffusion can travel with unbounded velocities, contradicting Einstein's principles of relativity. On the other hand, **diffusion** (mathematical) can be seen as **limit of the damped wave equation** with infinite damping and infinite propagation speed.

Let us consider the damped wave equation!

- ◆ What other physical PDEs can be used in generative AIs?
 - Xu, Liu, Tegmark, Jakkola, Poisson flows, NeuRips 2022
 - Liu, Lu, Xu, Jaakkola, Tengmark, GenPhys: from physical processes to generative models, 2023

Refs: Duong, Chemseddine, Friz, St., Telegraphers generative models via Kac flows, 2025

Last week follow up paper: DestillKac, Ermon et al. 2025

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Kac Process (Mark Kac 1974)

Particle starts at $x_0 = 0$ to the right and moves with step Δt and velocity $c > 0$ in either positive or negative direction, where

- ◆ same direction with probability $1 - a\Delta t$
- ◆ reverse direction with the probability $a\Delta t \rightarrow$ random variable $\varepsilon_k \in \{\pm 1\}$

Particle displacement S_n after n steps (random variable):

$$S_n = c\Delta t(1 + \varepsilon_1 + \varepsilon_1\varepsilon_2 + \dots + \varepsilon_1 \dots \varepsilon_n) = c\Delta t \sum_{k=0}^n (-1)^{N_k}$$

where $N_k \sim B(k, a\Delta t)$ number of reversals up to step k

- ◆ Continuous limit for $n \rightarrow \infty$ such that $na\Delta t \rightarrow at$, then $N_n \rightarrow \text{Poi}(at)$

Number of reversals $N(t)$ up to time t is given by a

homogeneous Poisson point process with rate a :

- $N(0) = 0$;
- Independent increments of $N(t)$
- $N(t) - N(s) \sim \text{Poi}(a(t - s))$ for all $0 \leq s < t$.

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Particle displacement:

$$S_t = c \int_0^t (-1)^{N(s)} ds$$

Kac process starting in 0: $K_t := B_{\frac{1}{2}} S_t, \quad B_{\frac{1}{2}} \sim B(1, \frac{1}{2})$

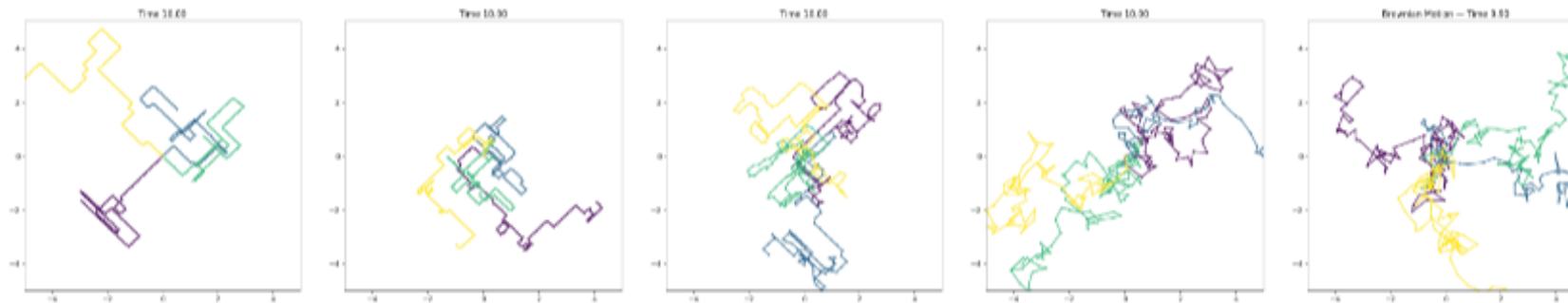


Figure 1: Paths of the componentwise Kac walk in 2D, simulated until time $T = 10$ with damping/velocity parameters $(a, c) = (1, 1), (2, 1), (4, 2), (25, 5)$, and a standard Brownian motion (right).

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Theorem. The **probability distribution** flow of $(K_t)_t$ is

$$\mu_t(x) = \frac{1}{2}e^{-at}(\delta_0(x+ct) + \delta_0(x-ct)) + \tilde{p}_t(x),$$

with the absolutely continuous part

$$\tilde{p}_t(x) := \frac{1}{2}e^{-at} \left(\beta ct \frac{I'_0(\beta r_t(x))}{r_t(x)} + \beta I_0(\beta r_t(x)) \right) 1_{[-ct, ct]}(x),$$

where

$$r_t(x) := \sqrt{c^2 t^2 - x^2}$$

and $\beta := \frac{a}{c}$, and I_0 is 0-th *modified Bessel function of first kind*. It is the generalized solution of the **damped wave equation/telegrapher equation**

$$\partial_{tt}u(t, x) + 2a \partial_t u(t, x) = c^2 \Delta_x u(t, x),$$

$$u(0, x) = \delta_0(x),$$

$$\partial_t u(0, x) = 0.$$

- $a > 0$ damping coefficient
- $c > 0$ velocity of the wave front

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Theorem. Let $f_0 = \delta_0$. Then μ_t associated to $u(t, \cdot)$ solves CE

$$\partial_t \mu_t = -\partial_x (\mu_t v_t)$$

in a weak sense with **velocity field**

$$v_t(x) := \begin{cases} \frac{x}{t + \frac{r_t(x)}{c} \frac{I_0(\beta r_t(x))}{I_0'(\beta r_t(x))}} & \text{if } x \in (-ct, ct), \\ c & \text{if } x = ct, \\ -c & \text{if } x = -ct, \\ \text{arbitrary} & \text{otherwise.} \end{cases}$$

It holds $\|v_t\|_{L_2(\mu_t)}^2 \leq c^2$, and hence $\|v_t\|_{L_2(\mu_t)} \in L_2(0, 1)$. 😊

- ◆ Compare to diffusion starting in δ_0 : $v_t(x) = \frac{x}{2t}$
- ◆ Damped wave equation is in **general no longer mass-conserving in dimension $d \geq 3$** Refs: Tautz, Lerche 2016

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Theorem (Kac Insertion Methods) Refs: Griego/Hersh 1971

For any $f_0 \in H^2(\mathbb{R}^d)$, let $w_c(t, x)$ solve the **undamped wave equation**

$$\begin{aligned}\partial_{tt}w(t, x) &= c^2 \Delta w(t, x), \quad x \in \mathbb{R}^d, \quad t > 0, \\ w(0, x) &= f_0(x), \quad \partial_t w(0, x) = 0.\end{aligned}$$

Then, the functions defined by

$$\begin{aligned}u(t, x) &:= \mathbb{E}[w_c(c^{-1}S_t, x)] \\ h(t, x) &:= \mathbb{E}[w_1(\sigma W_t, x)],\end{aligned}$$

multi-dimensional **damped wave equation**, resp. **the heat equation**

$$\begin{aligned}\partial_t h(t, x) &= \frac{\sigma^2}{2} \Delta h(t, x), \quad x \in \mathbb{R}^d, \quad t > 0, \\ h(0, x) &= f_0(x).\end{aligned}$$

Corollary. The solution $u^{a,c}(t, \cdot)$ of the damped heat equation converges to the solution $h(t, \cdot)$ of the diffusion equation for $a, c \rightarrow \infty$ with fixed $\sigma^2 = \frac{c^2}{a}$.

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Numerical Example

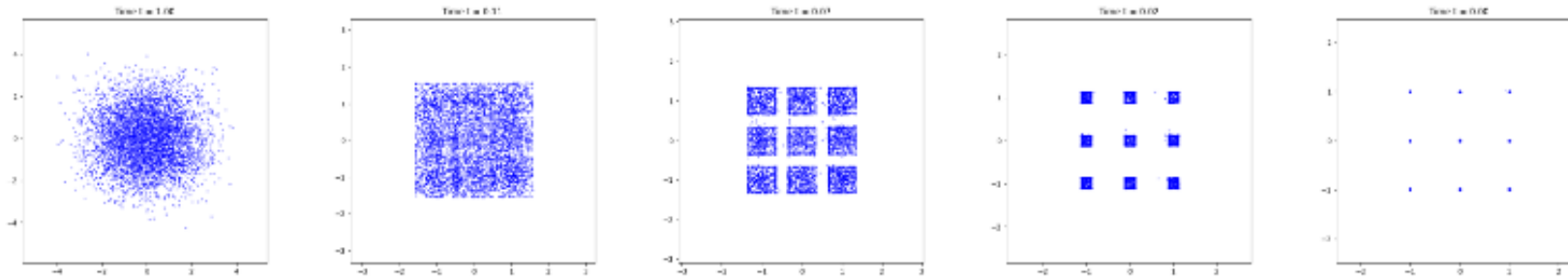


Figure 4: Backward evolution of the *learned* Kac flow for $(a, c) = (25, 5)$, see also Figure 5.

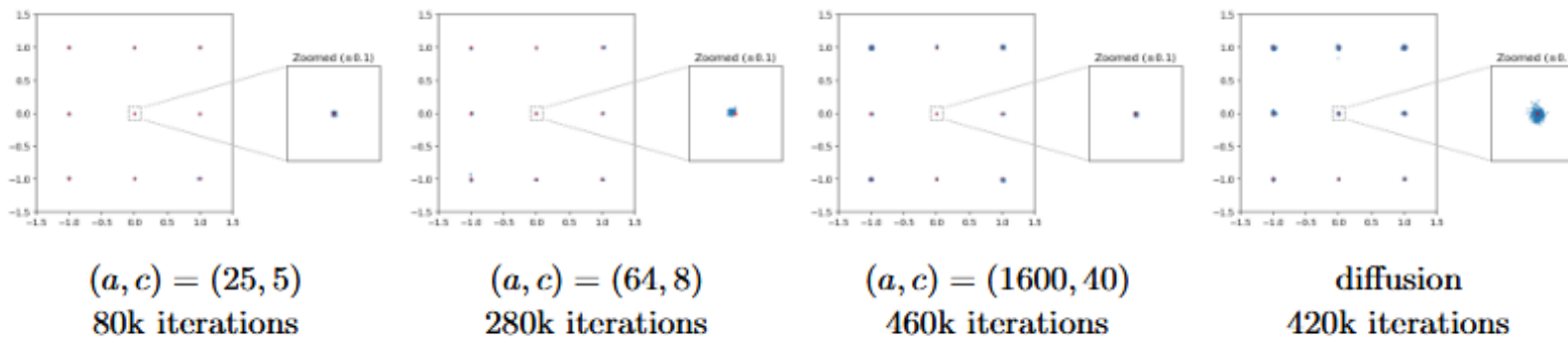
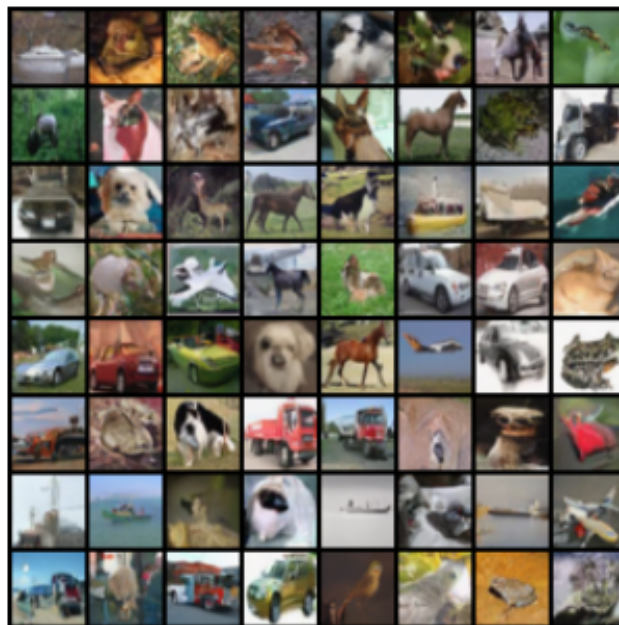


Figure 5: Generated samples (blue) vs. ground truth (red) at the indicated iteration for each model. The Kac model can precisely recover the small modes, while the diffusion model creates "blobs".

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Numerical Example

Schedule: $g(t) = t^2$			
Method	FID	Method	FID
$a = 900, c = 10$	7.26	$a = 100, c = 10$	10.01
$a = 900, c = 20$	7.46	$a = 25, c = 1$	8.60
$a = 900, c = 30$	8.05	$a = 25, c = 2$	8.65
$a = 100, c = 1$	11.41	$a = 25, c = 3$	9.15
$a = 100, c = 3$	7.77	$a = 25, c = 4$	9.56
$a = 100, c = 5$	7.73	$a = 25, c = 5$	10.70
$a = 100, c = 7$	8.58	FM (our impl.)	7.59



$(a, c) = (900, 10)$

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3. Uniform Process and MMD Gradient Flow

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Gradient flow of the maximum mean discrepancy functional $\mathcal{F}_\nu = \text{MMD}_K(\cdot, \nu)$ with negative distance kernel $K(x, y) = -|x - y|$ and $\nu = \mathcal{U}(-b, b)$, i.e. (μ_t, v_t) fulfill the continuity equation with

$$v_t \in -\partial \mathcal{F}_\nu(\mu_t)$$

Theorem The Wasserstein gradient flow μ_t of $\mathcal{F}_\nu$ starting in $\mu_0 = \delta_0$ towards the uniform distribution $\nu = \mathcal{U}[-b, b]$ is

$$\mu_t = (1 - \exp(-\frac{t}{b}))\mathcal{U}[-b, b], \quad t > 0,$$

with corresponding velocity field

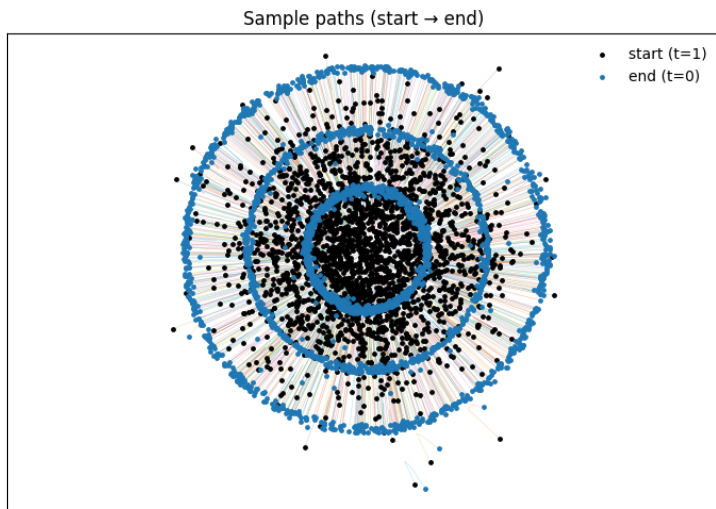
$$v_t(x) = \frac{x}{b \left(\exp\left(\frac{t}{b}\right) - 1 \right)}, \quad x \in \text{supp}(\mu_t).$$

A corresponding (stochastic) process $(U_t)_t$ is given by

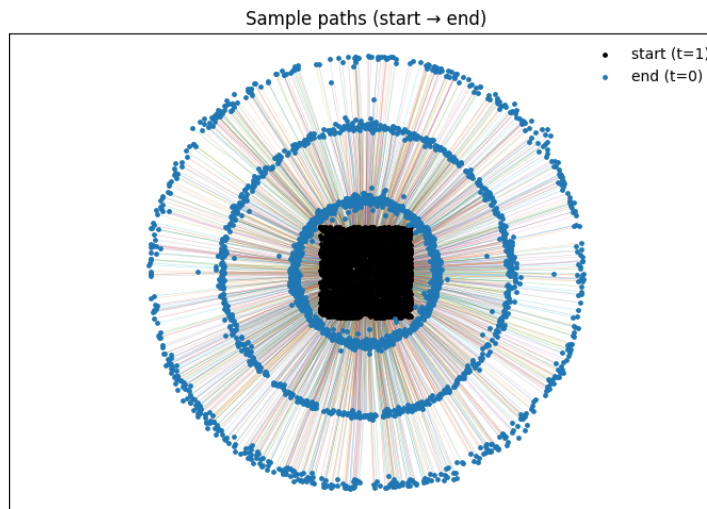
$$U_t := b \left(1 - \exp\left(-\frac{t}{b}\right) \right) U, \quad U \sim \mathcal{U}[-1, 1].$$

It holds $\|v_t\|_{L_2(\mu_t)}^2 = \frac{2b}{3} \exp(-\frac{2t}{b})$, and hence $\|v_t\|_{L_2(\mu_t)} \in L_2(0, 1)$. 😊

Example



IMM + Gaussian



IMM + MMD

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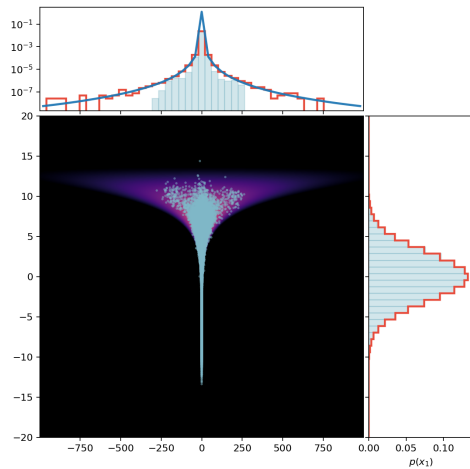
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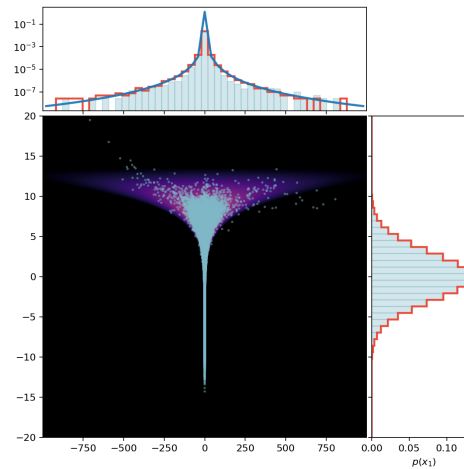
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3.2 Learned Processes

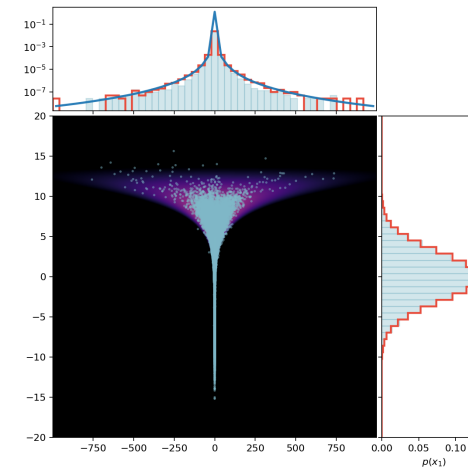
- ◆ Besides exploding velocity fields, starting at a Gaussian may encounter difficulties if e.g.
the target is **heavy tailed or multimodal or ...**



Gaussian



Student-T



Our (Learned)

Refs:

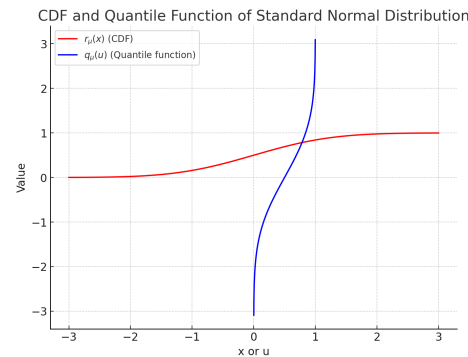
Hagemann/Neumayer, Stabilizing invertible neural networks using mixture models, Inverse Problems 2021,
Salomea et al., Can push-forward generative models fit multimodal distributions?, Neurips 2022

Pandey et al., Heavy-tailed diffusion models, 2024

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◆ **Cumulative distribution function:** $R_\mu(x) := \mu((-\infty, x]), x \in \mathbb{R}$

◆ **Quantile function:** $Q_\mu(u) := \min\{x \in \mathbb{R} : R_\mu(x) \geq u\}, u \in (0, 1)$



◆ Quantile functions form a closed, convex cone in $L_2(0, 1)$

$$\mathcal{C} := \{f \in L_2(0, 1) : f \text{ increasing a.e.}\}$$

◆ $\mu \mapsto Q_\mu$ is an isometric **embedding** of $(\mathcal{P}_2(\mathbb{R}), W_2)$ into $(L_2(0, 1), \|\cdot\|_{L_2})$:

$$W_2^2(\mu, \nu) = \int_0^1 |Q_\mu(s) - Q_\nu(s)|^2 ds$$

and $Q_\mu \circ U \sim \mu$, resp. $\mu = Q_{\mu, \#} \mathcal{L}_{(0,1)}$

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Idea: Learn quantile functions $\mathbf{Q}_\phi = (Q_\phi^i)_{i=1}^d$ taking their monotonicity into account Refs. monotonicity parameterization: Durkan et al. 2019 and consider

$$\mathbf{X}_t = (1 - t)\mathbf{X}_0 + t\mathbf{Q}_\phi(\mathbf{U}) \quad \text{with i.i.d. } U^i \sim \mathcal{U}[0, 1]$$

such that the noise $\mathbf{Y}_1 = (Q_\phi^i(U^i))_{i=1}^d$ adapts to the given data from $\mathbf{X}_0$.
By construction

$$\text{Law}(\mathbf{Y}_1) \in \mathcal{S} := \{\nu \in \mathcal{P}_2(\mathbb{R}^d) : \nu = \rho \# \mathbf{x} \text{ and } \rho = \Pi_{i=1}^d \rho^i\},$$

so that it appears reasonable to ask for

$$\mathcal{E}(\phi) = W_2^2(\mu_0, \nu_\phi), \quad \nu_\phi := (\mathbf{Q}_\phi)_\# \mathcal{U}([0, 1]^d).$$

Loss Function: Take also the velocity field into account

$$\mathcal{L}(\theta, \phi) = \mathcal{E}_{\text{CFM}}(\theta, \phi) + \lambda \mathcal{E}(\phi), \quad \lambda > 0,$$

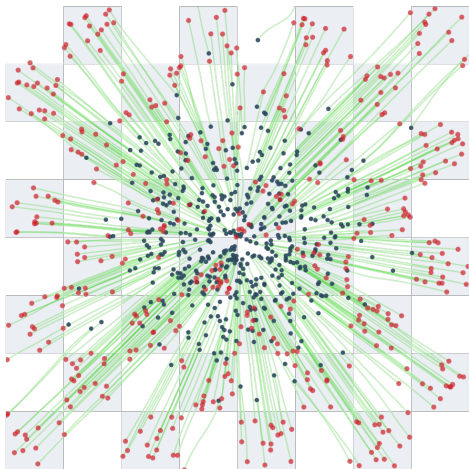
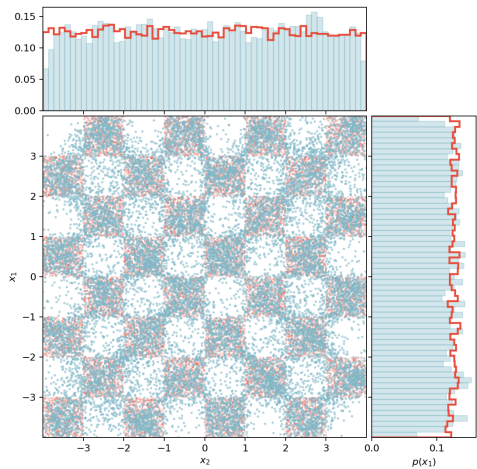
$$\mathcal{E}_{\text{CFM}}(\theta, \phi) = \mathbb{E}_{t \sim \mathcal{U}(0,1), (x_0, x) \sim \pi_\phi} \left[\left\| v_\theta((1-t)x_0 + tx, t) - (x_0 - x) \right\|_2^2 \right],$$

where $\pi_\phi \in \Pi_o(\mu_0, \nu_\phi)$

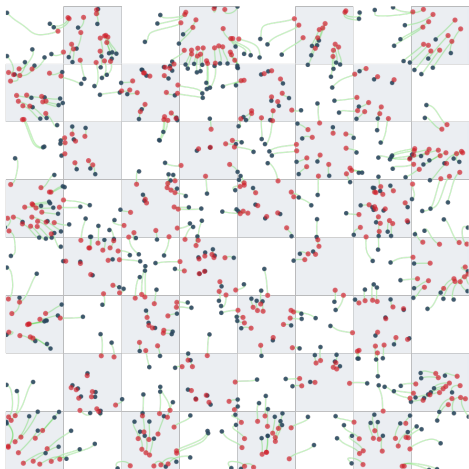
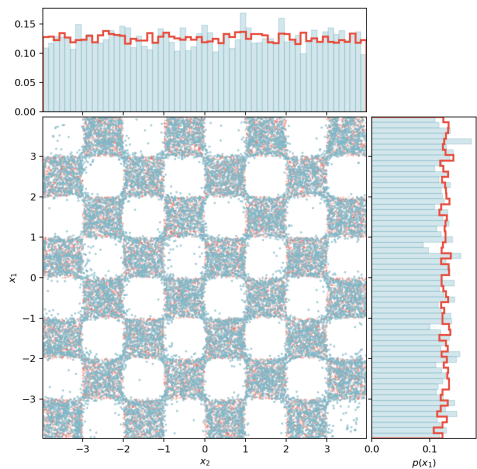
1	2
3	4
5	6
7	8
9	10
11	12
13	14
15	16
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19	20
21	22
23	24
25	26
27	28

Example: Checkerboard

Gaussian:



Learned:

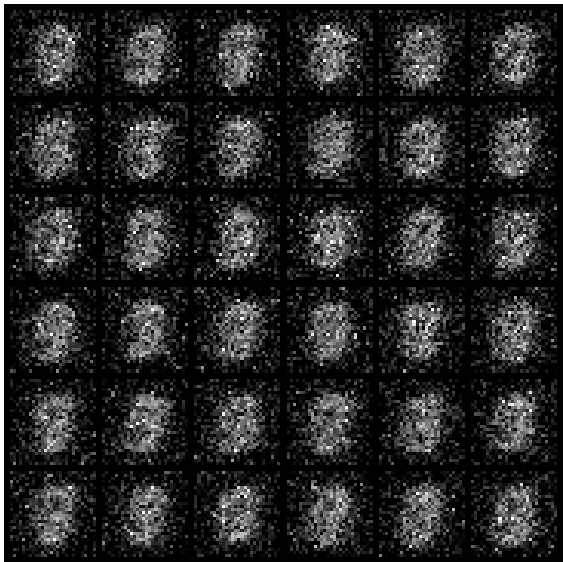


Noise:



1	2
3	4
5	6
7	8
9	10
11	12
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15	16
17	18
19	20
21	22
23	24
25	26
27	28

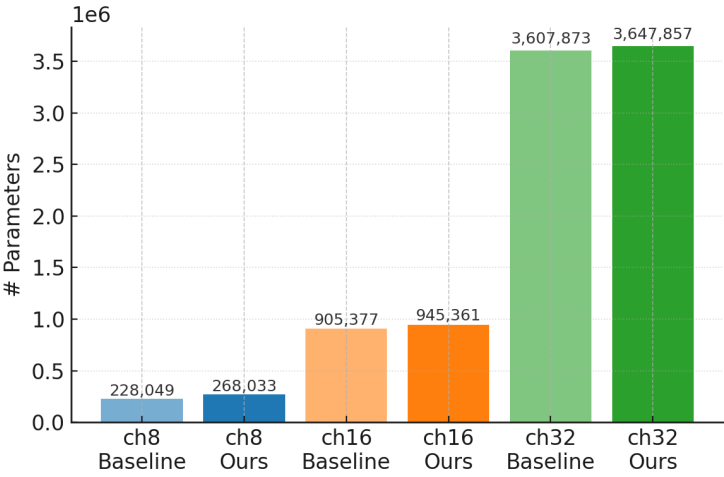
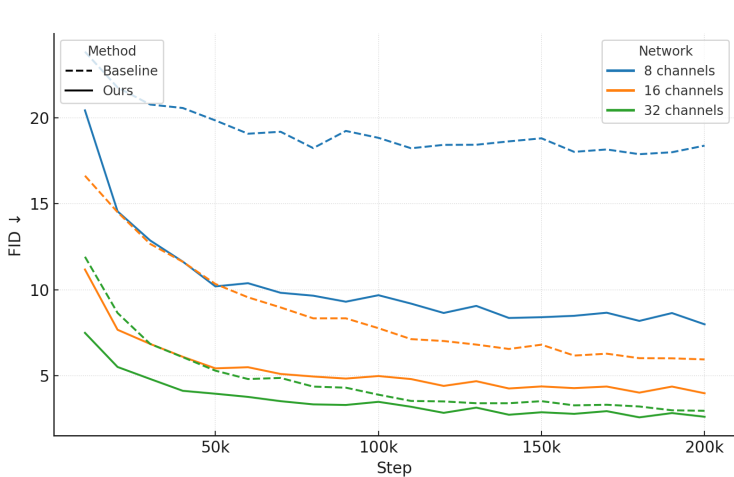
Example: MNIST



Samples from the learned latent



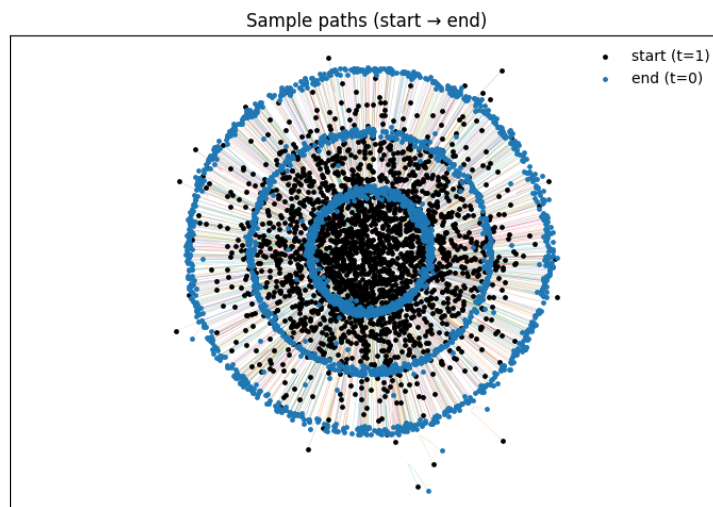
Generated samples



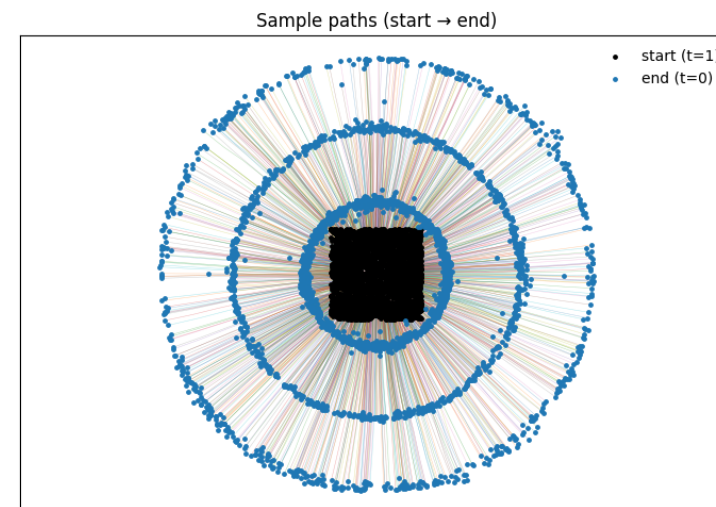
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17	18
19	20
21	22
23	24
25	26
27	28

3. Conclusions

- ◆ "Quantile sandbox" for building generative models: a unifying theory and a practical toolkit that turns noise selection into a data-driven design element
- ◆ Seamless fit into standard objectives including Flow Matching and consistency models, e.g. Inductive Moment Matching (IMM) via so-called **quantile interpolants**



IMM + Gaussian

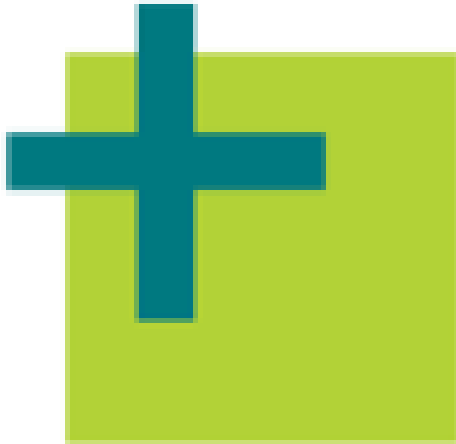


IMM + MMD

- ◆ Promising directions for future research
- ◆ References:
 1. Duong, Chemseddine, Friz, St: Telegrapher's Generative Model via Kac Flows, ArXiv 2025
 2. Chemseddine, Kornhardt, Duong, St: Adapting data to noise: generative flows from 1D processes, ArXiv this week

1	2
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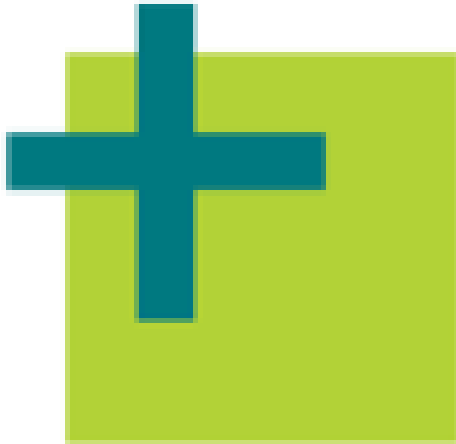








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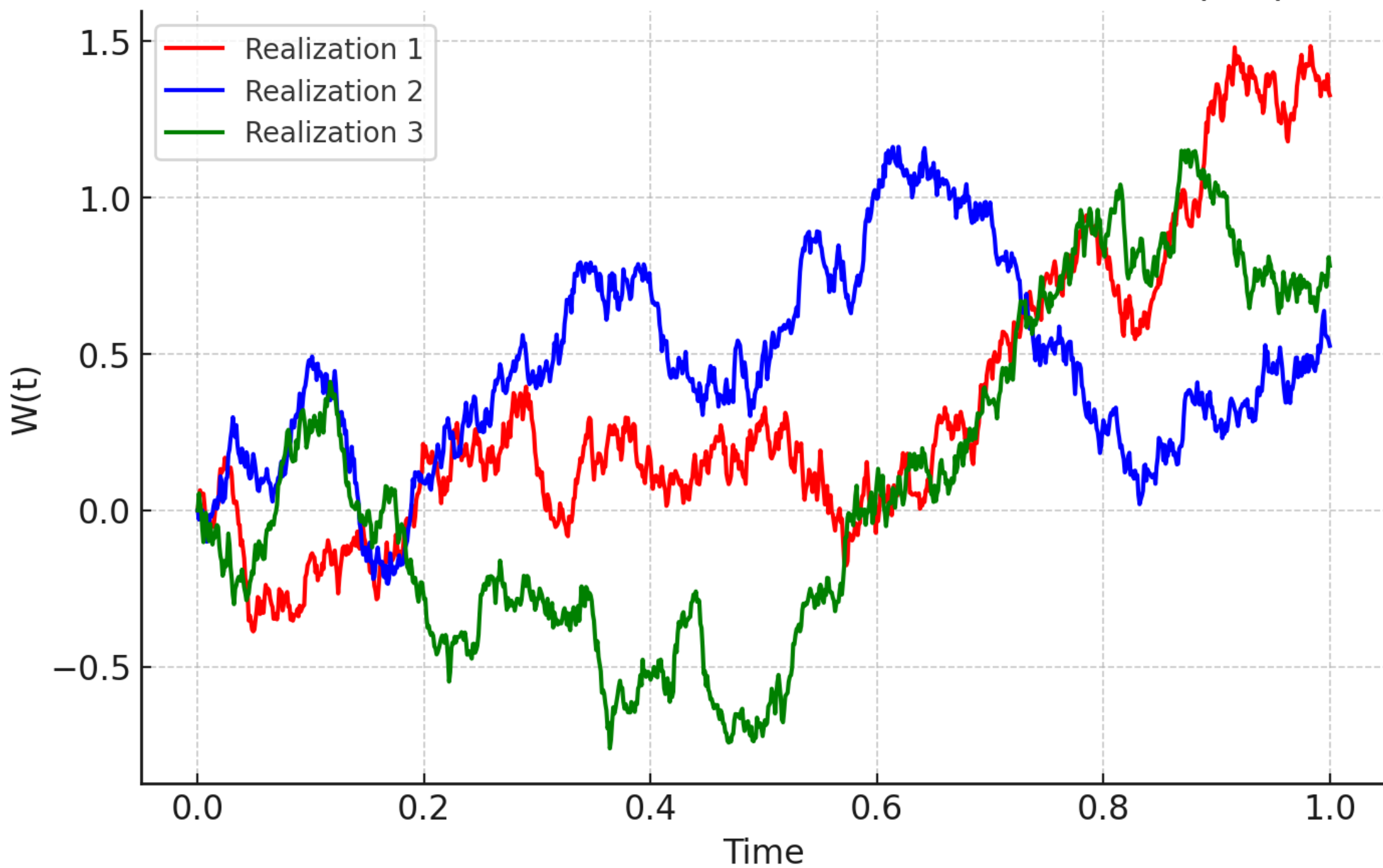
Theoretical Foundations of Deep Learning

DFG-funded Priority Program 2298

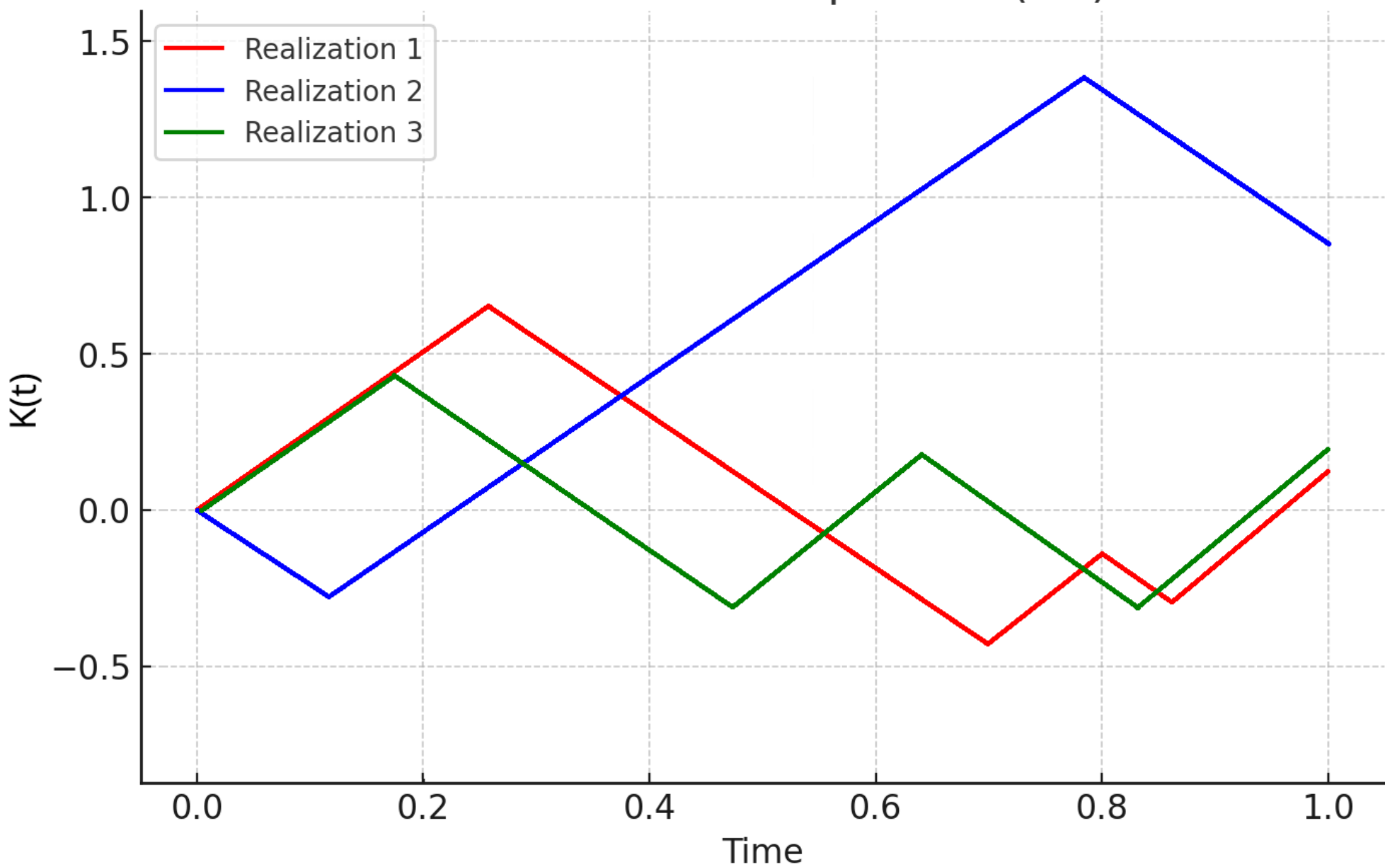
Funded under Germany's Excellence Strategy by
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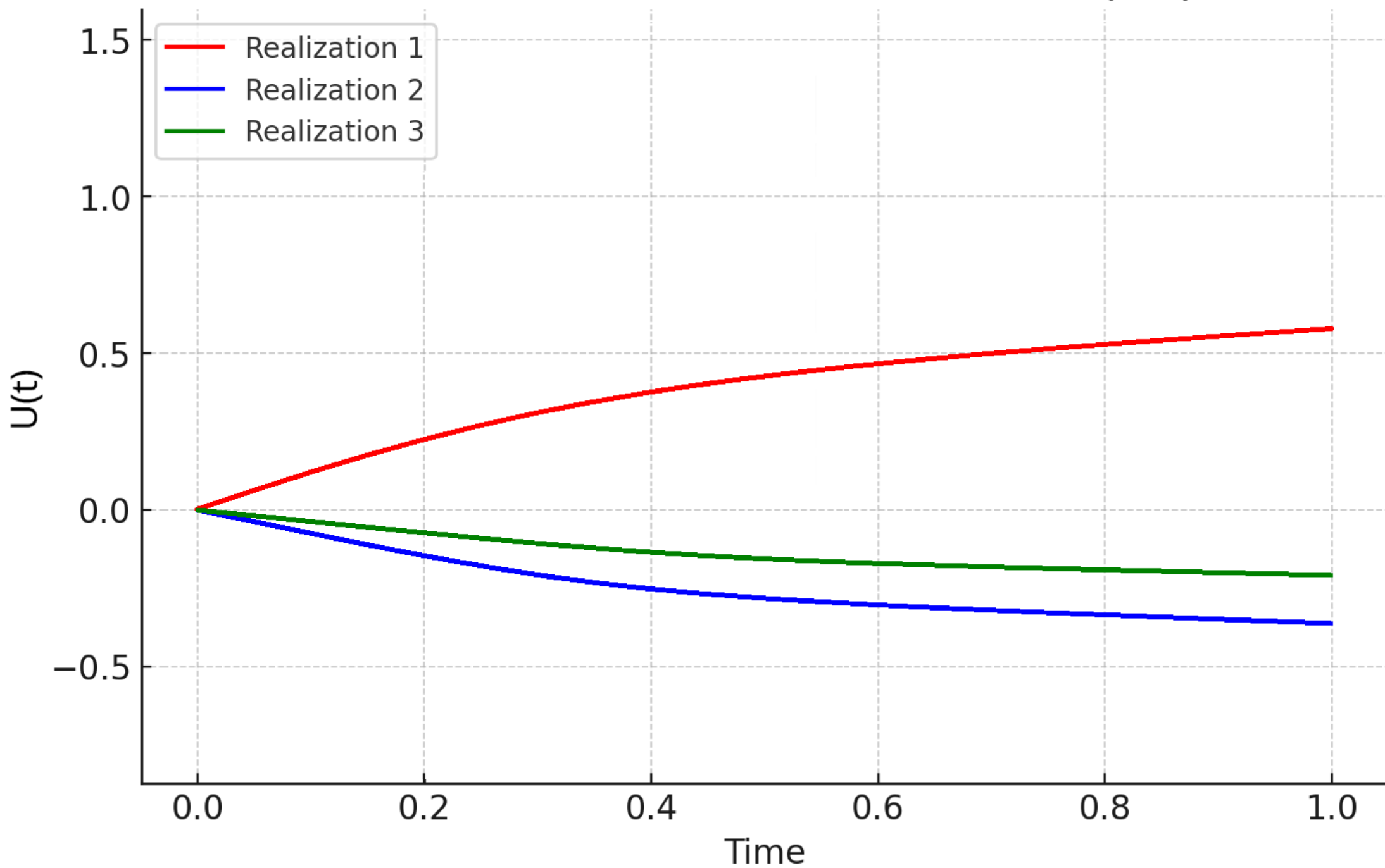
Realisations of a Standard Wiener Process (1D)

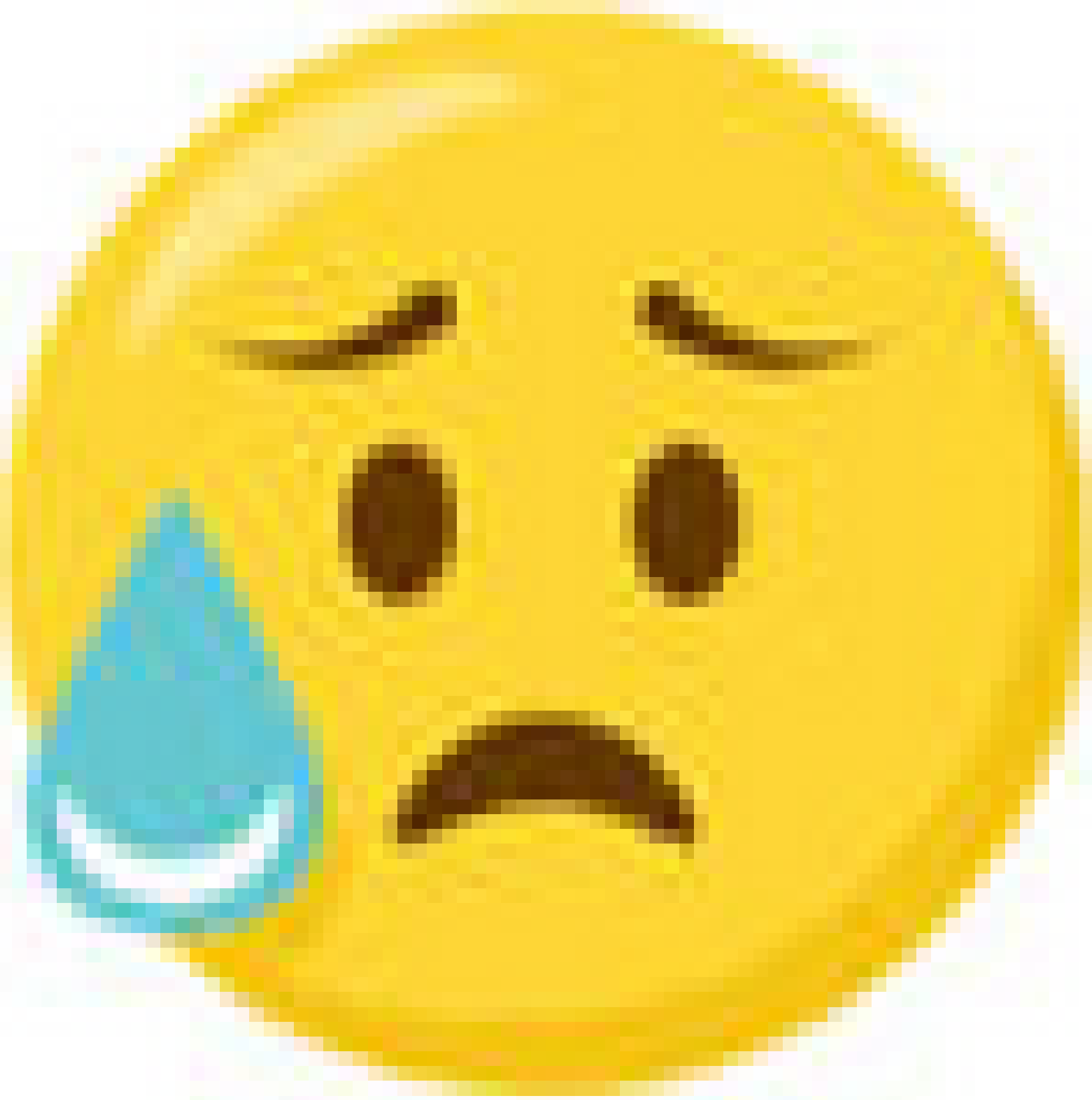


Realisations of the Kac process (1D)



Realisations of the Uniform Process (1D)





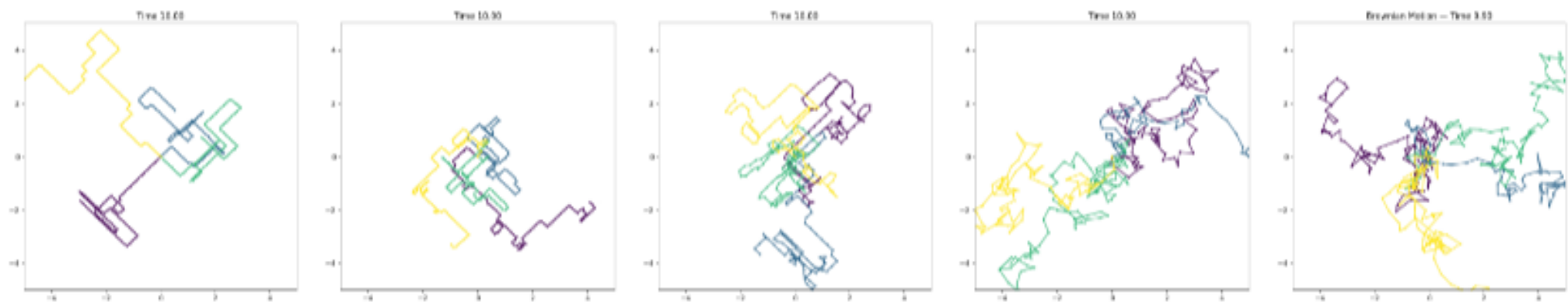


Figure 1: Paths of the componentwise Kac walk in 2D, simulated until time $T = 10$ with damping/velocity parameters $(a, c) = (1, 1), (2, 1), (4, 2), (25, 5)$, and a standard Brownian motion (right).



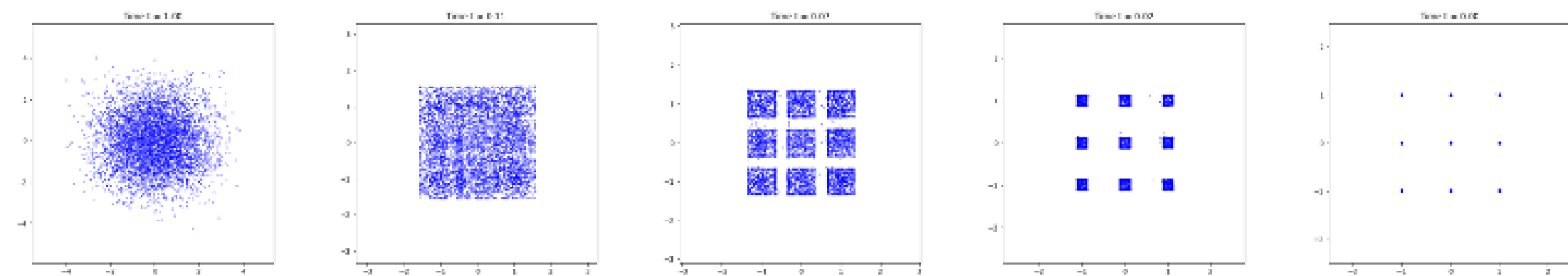


Figure 4: Backward evolution of the *learned* Kac flow for $(a, c) = (25, 5)$, see also Figure [5](#).

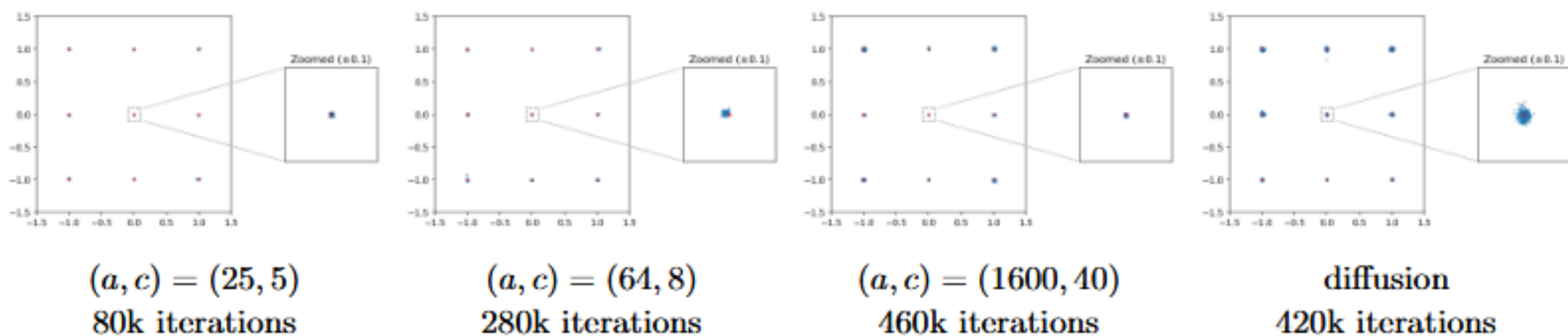
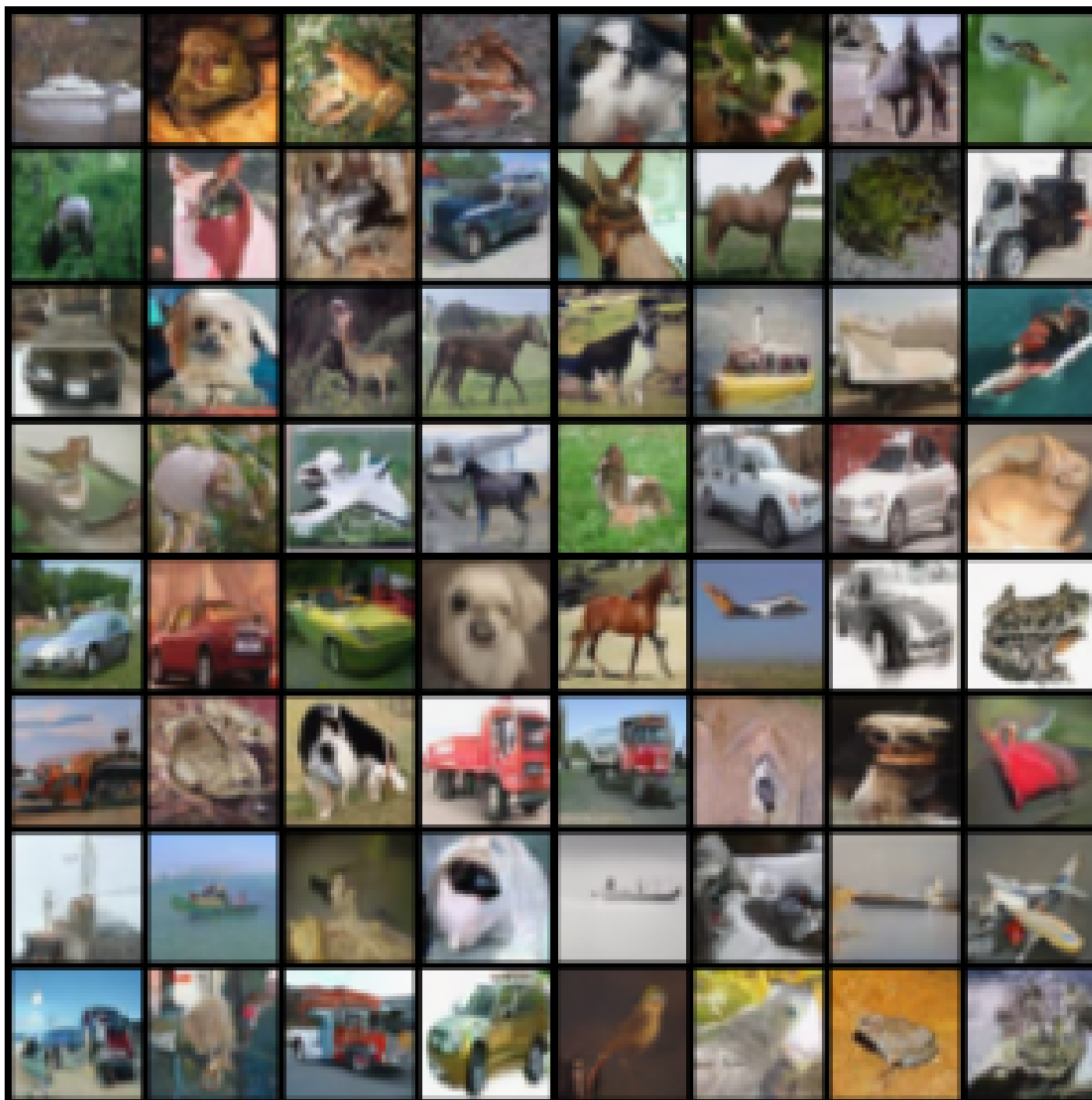


Figure 5: Generated samples (blue) vs. ground truth (red) at the indicated iteration for each model. The Kac model can precisely recover the small modes, while the diffusion model creates "blobs".

Schedule: $g(t) = t^2$

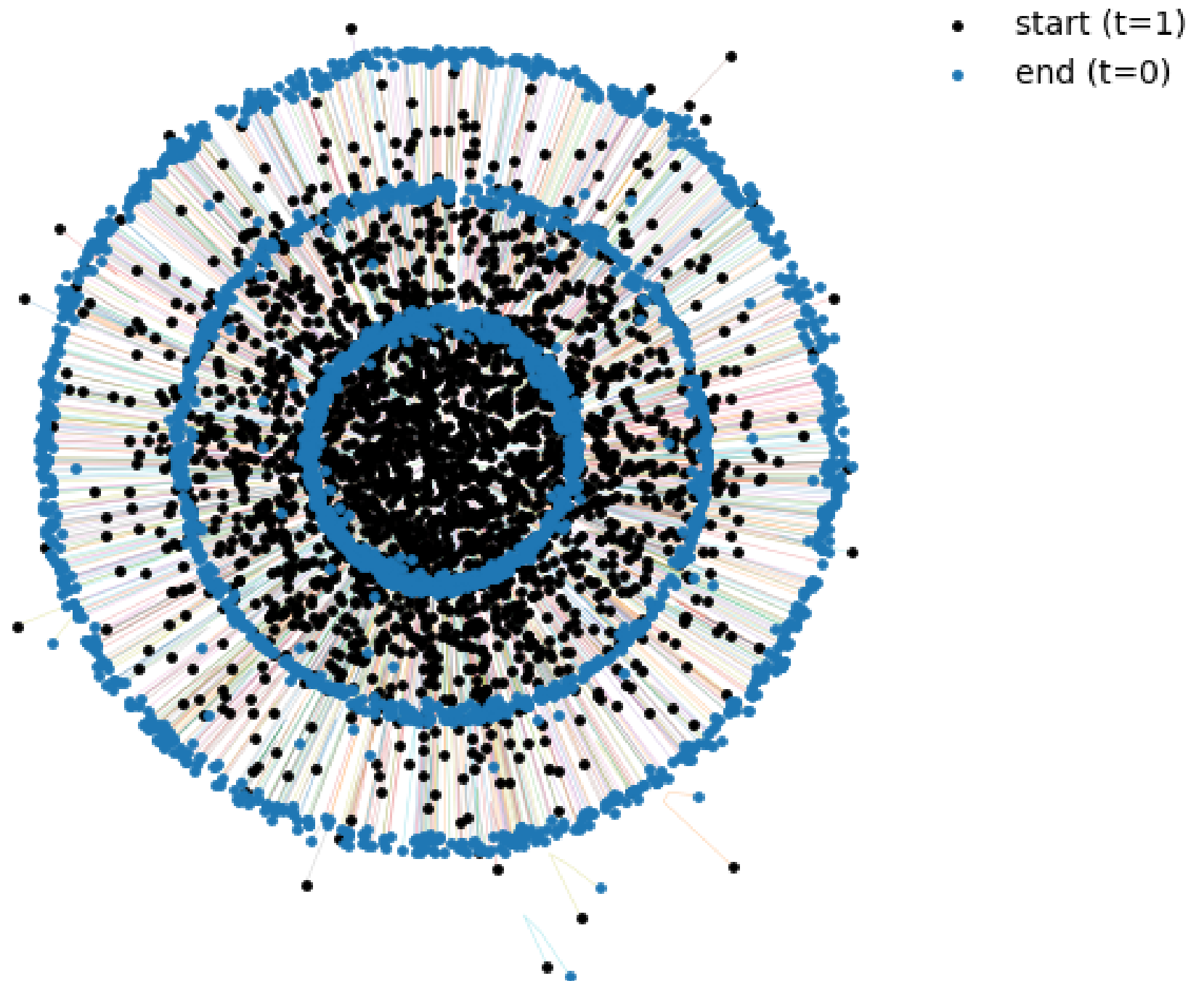
Method	FID	Method	FID
$a = 900, c = 10$	7.26	$a = 100, c = 10$	10.01
$a = 900, c = 20$	7.46	$a = 25, c = 1$	8.60
$a = 900, c = 30$	8.05	$a = 25, c = 2$	8.65
$a = 100, c = 1$	11.41	$a = 25, c = 3$	9.15
$a = 100, c = 3$	7.77	$a = 25, c = 4$	9.56
$a = 100, c = 5$	7.73	$a = 25, c = 5$	10.70
$a = 100, c = 7$	8.58	FM (our impl.)	7.59



$$(a, c) = (900, 10)$$

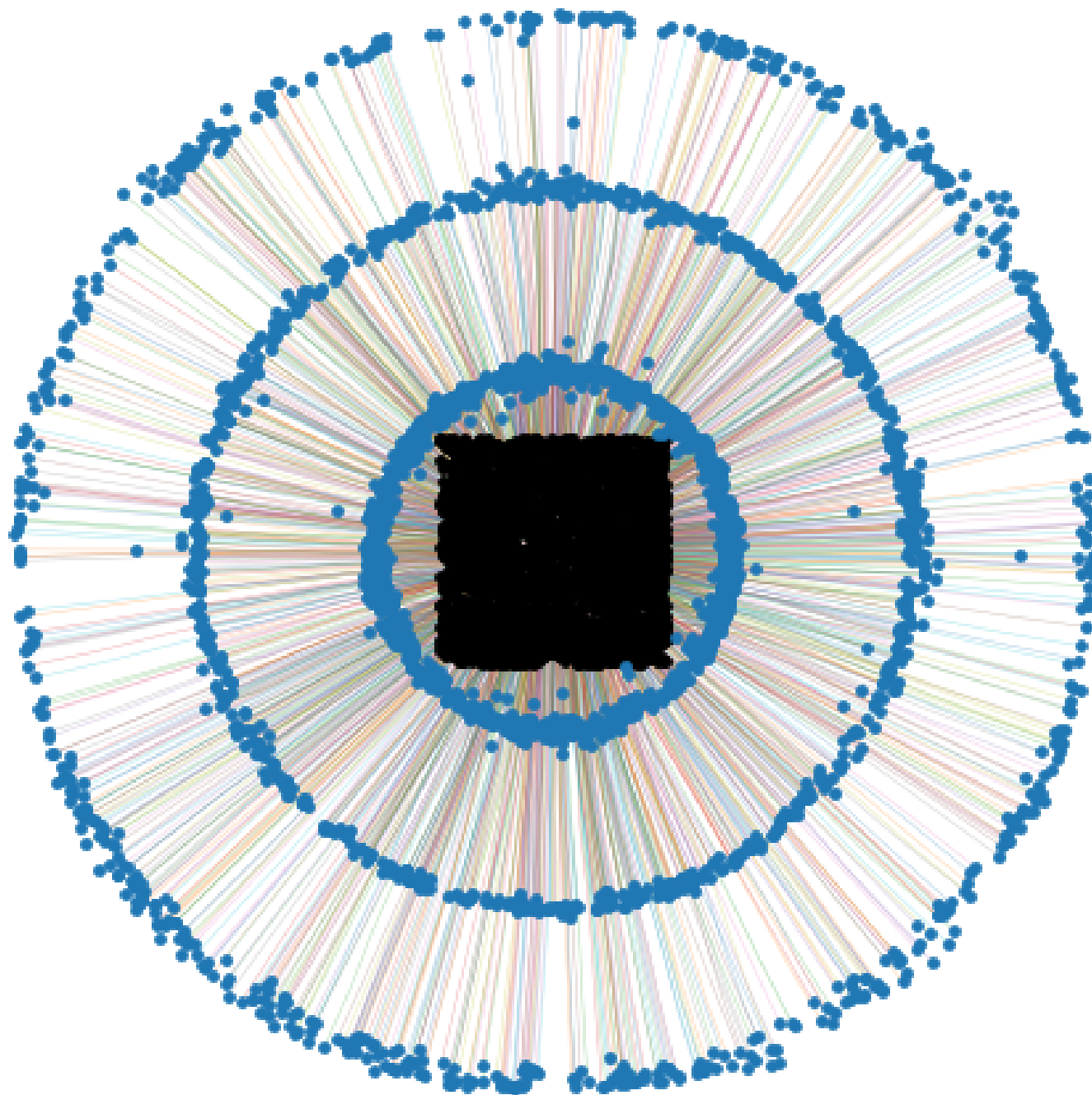


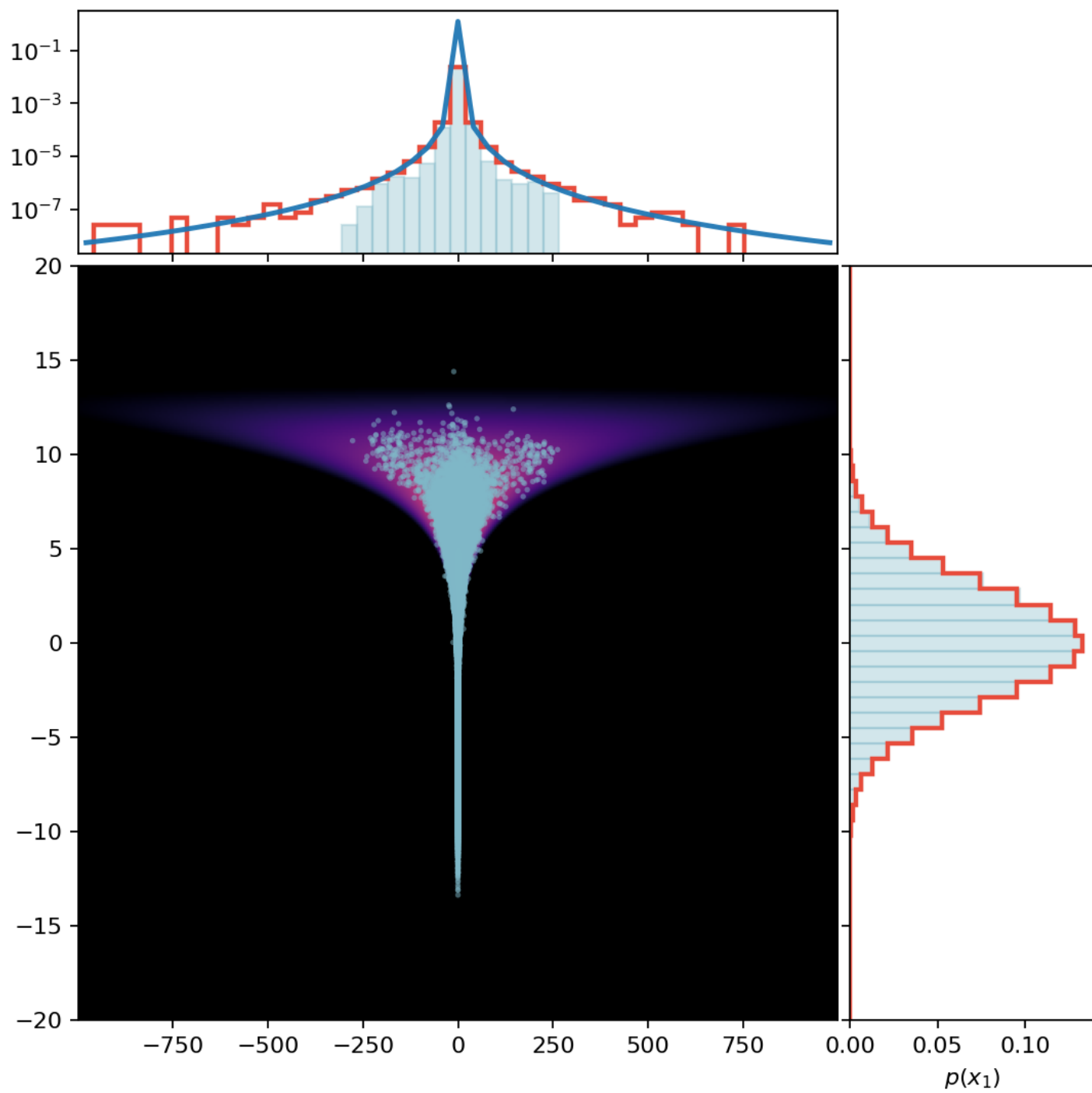
Sample paths (start $\rightarrow$ end)

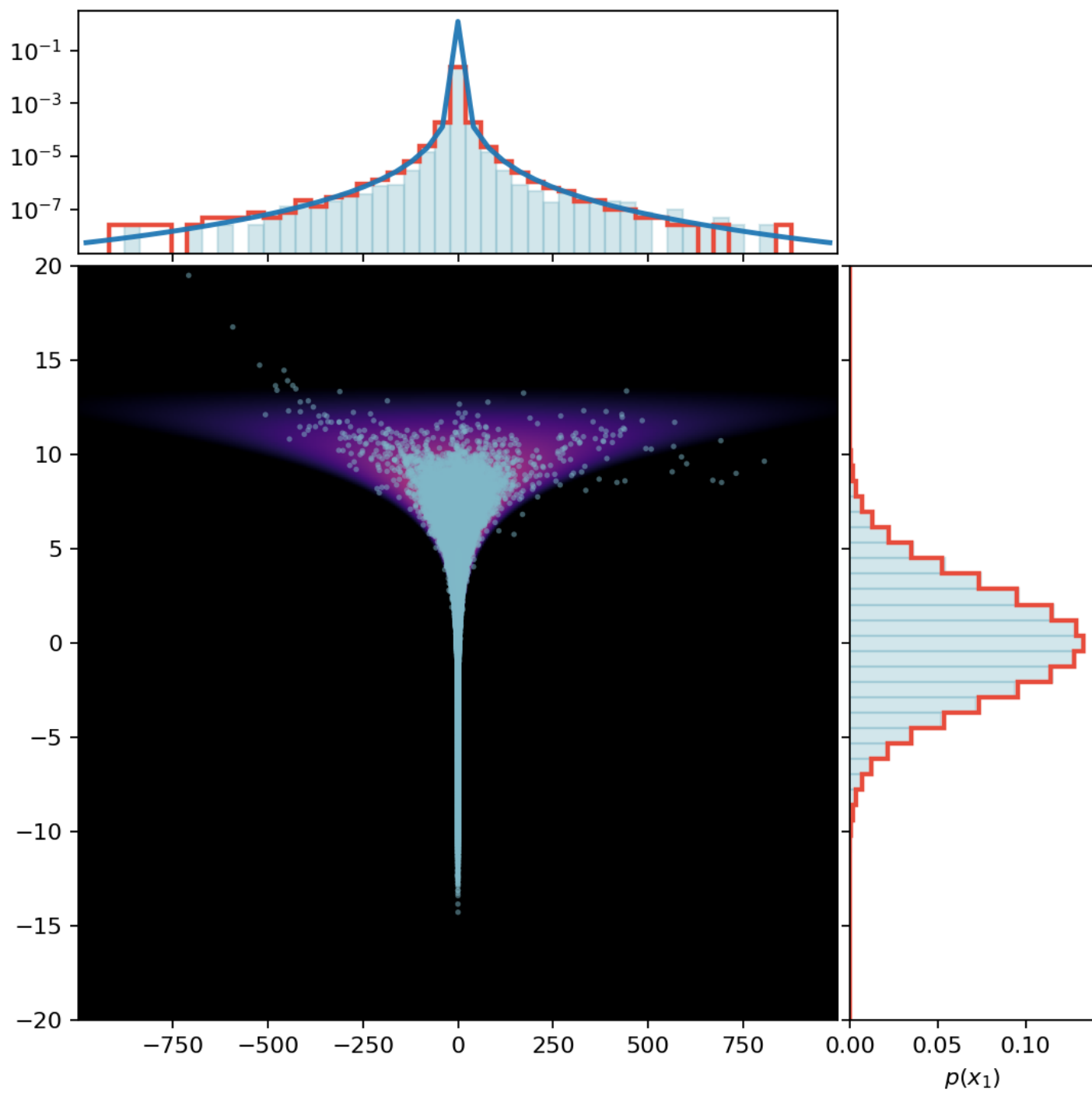


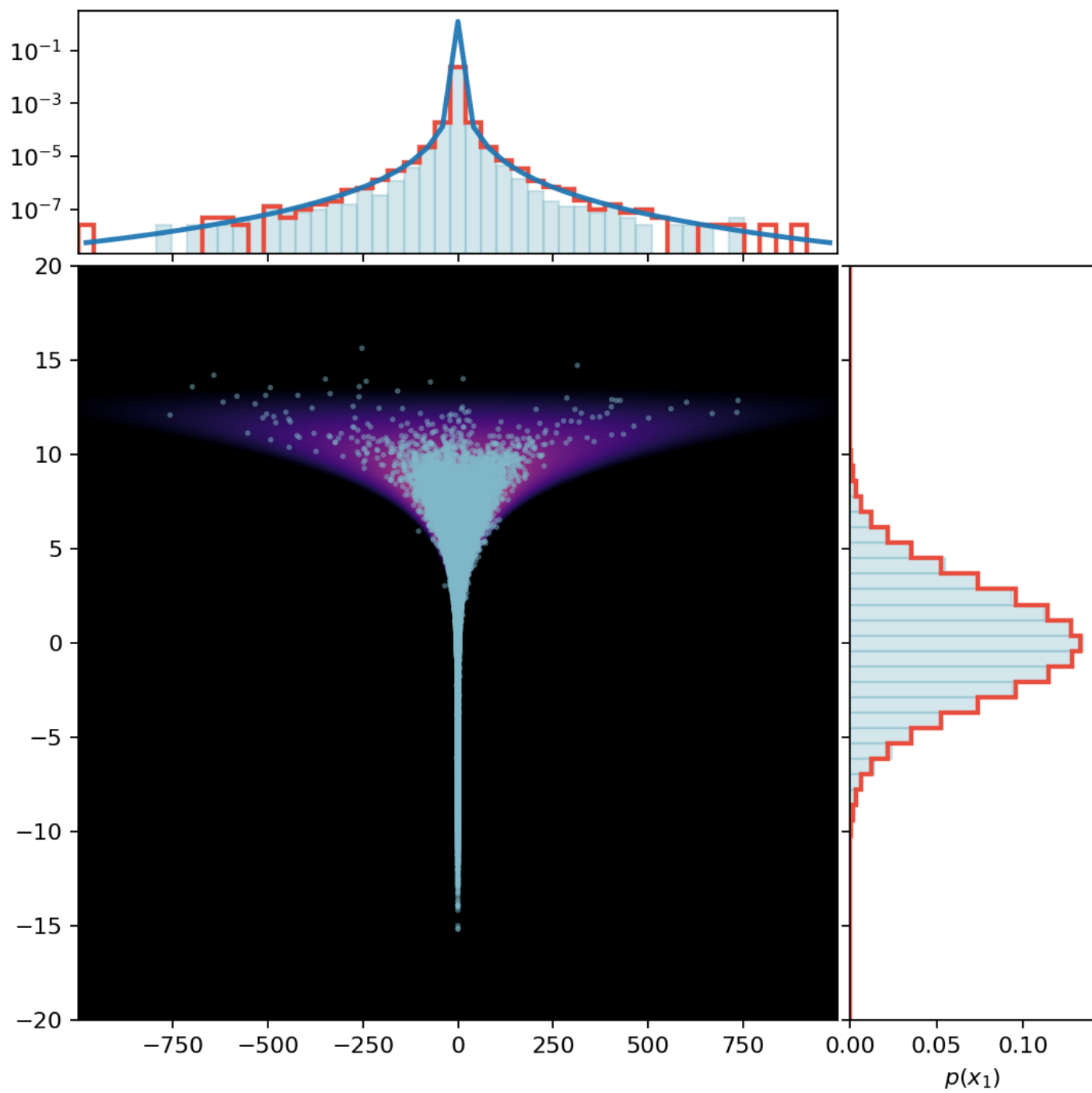
Sample paths (start $\rightarrow$ end)

- start ($t=1$)
- end ($t=0$)

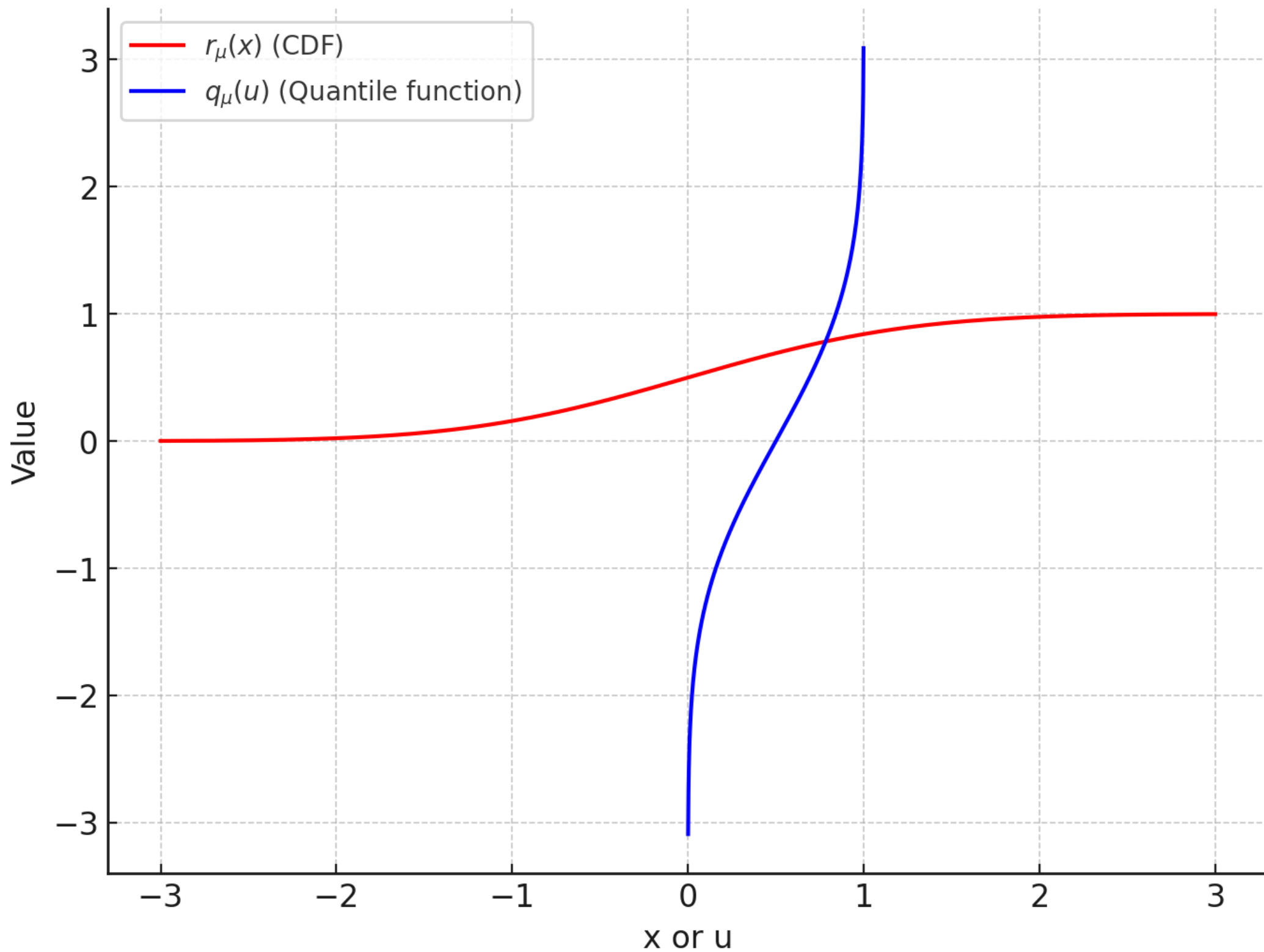


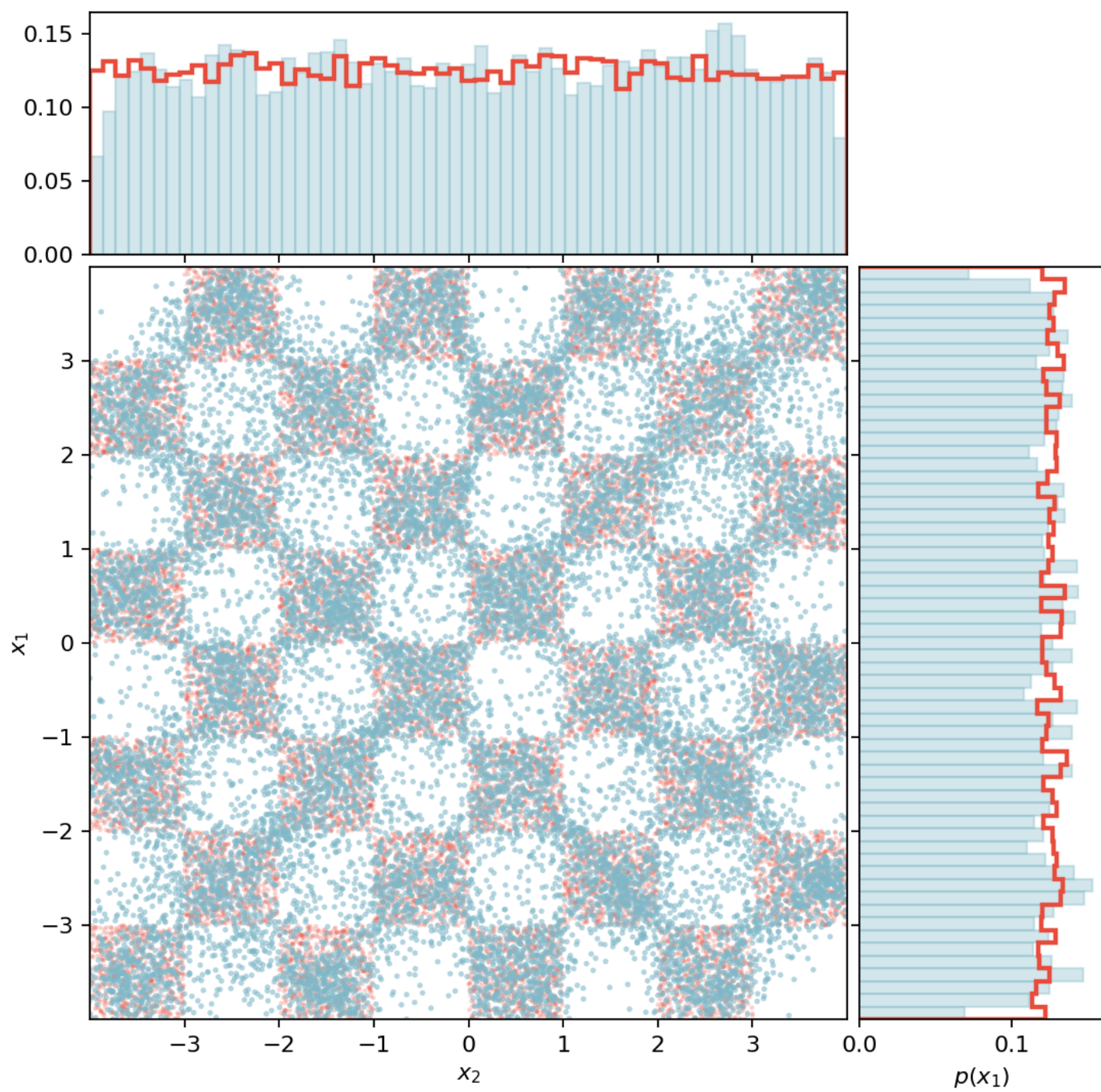


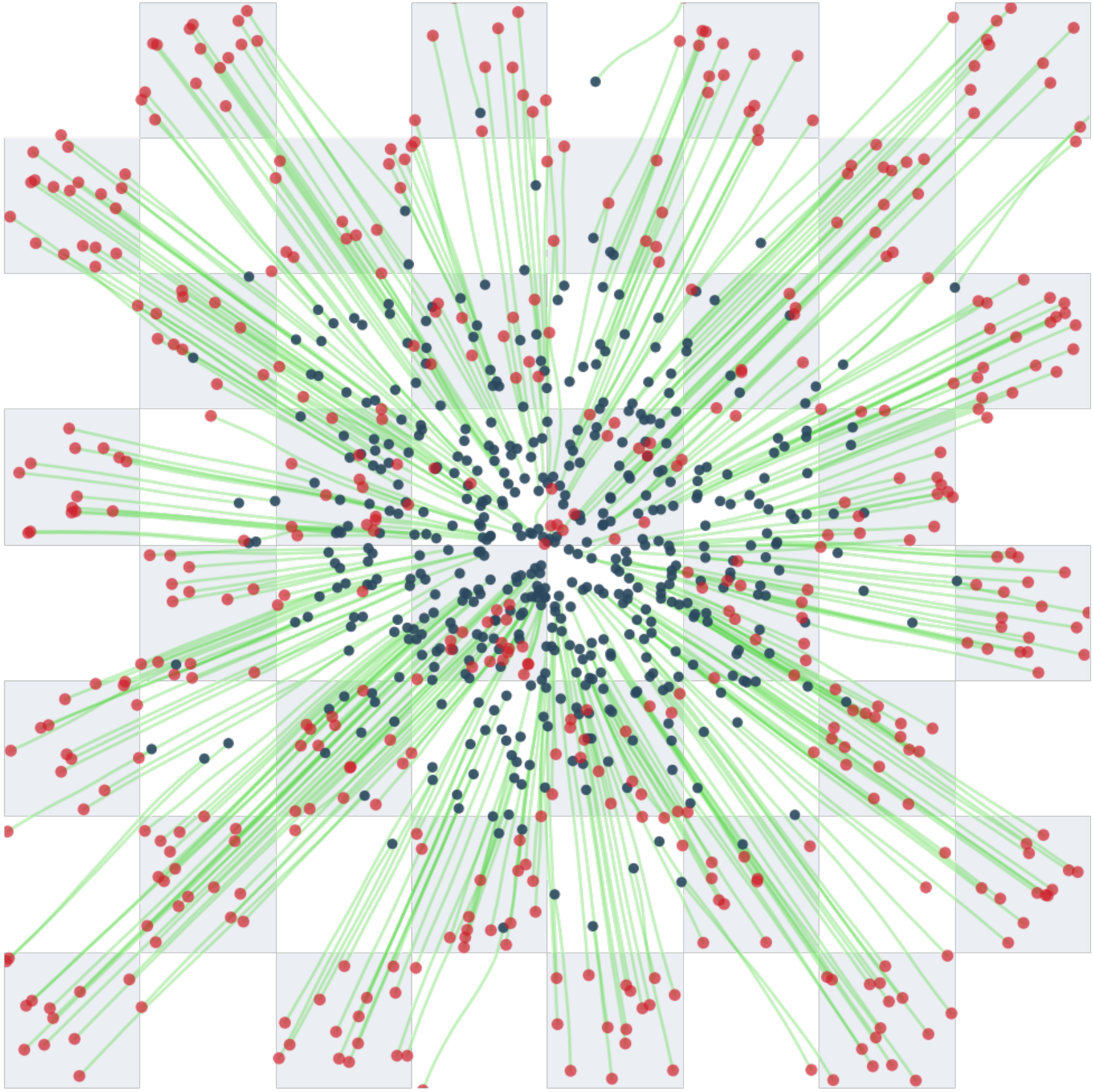


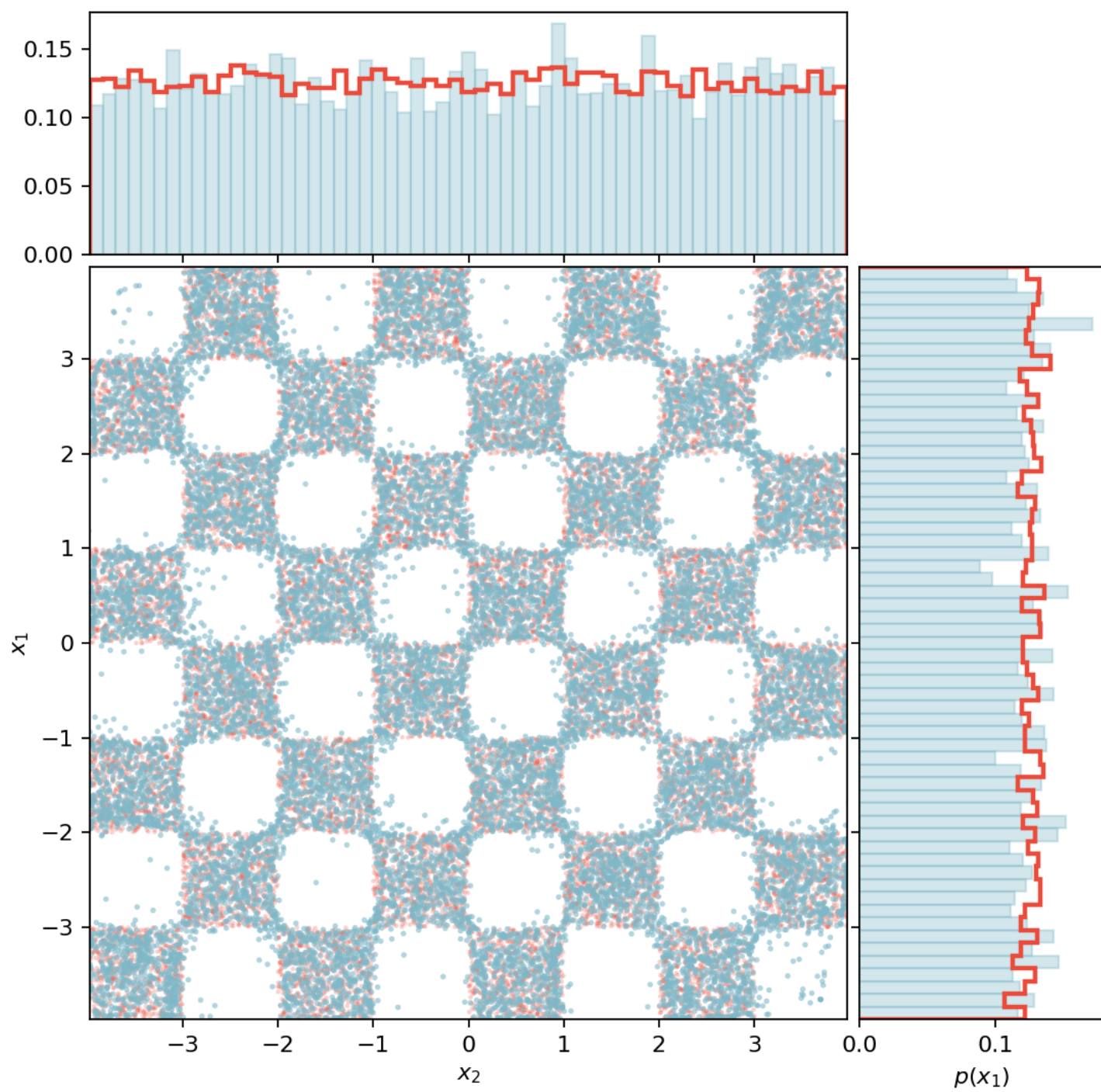


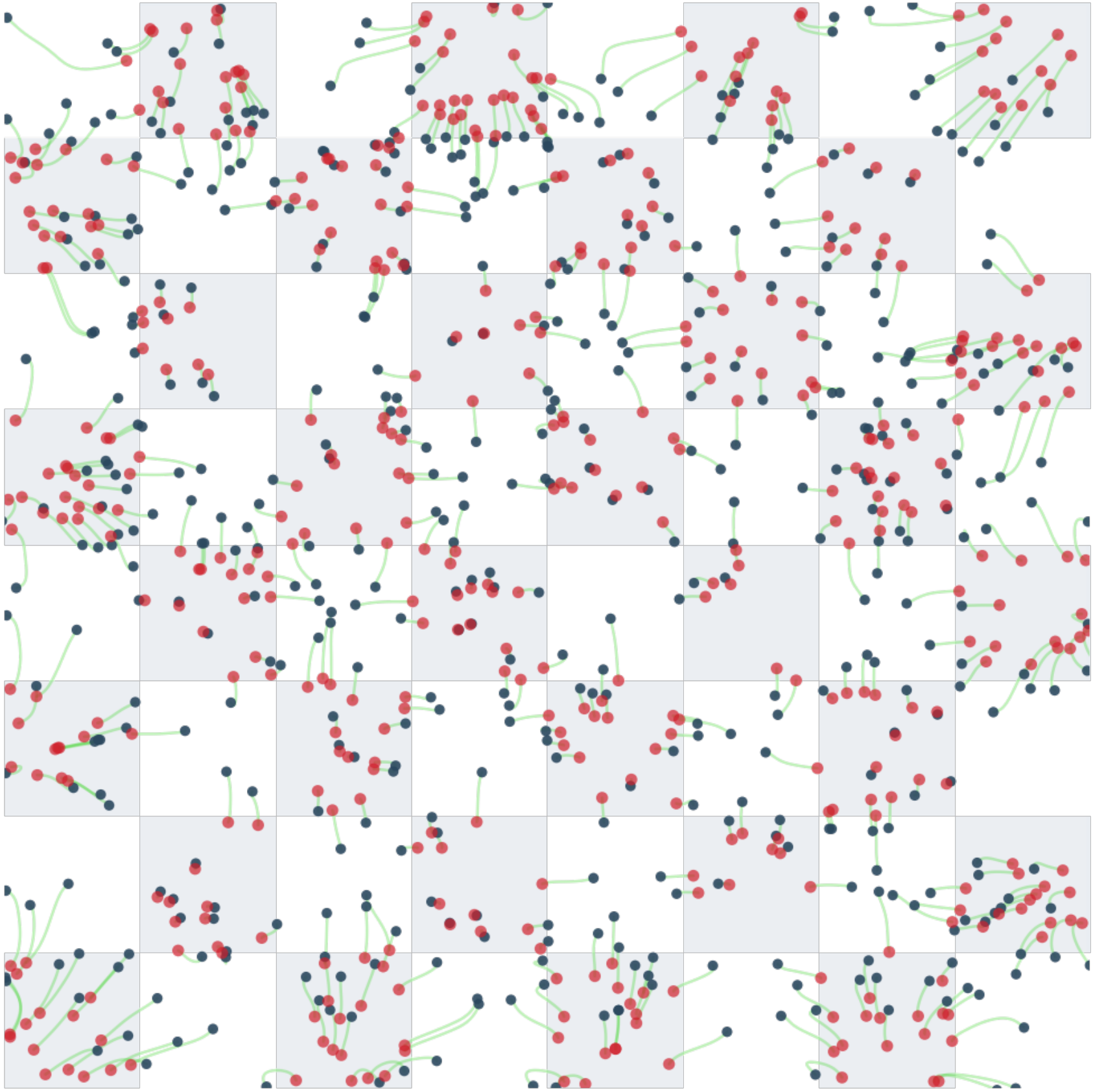
CDF and Quantile Function of Standard Normal Distribution

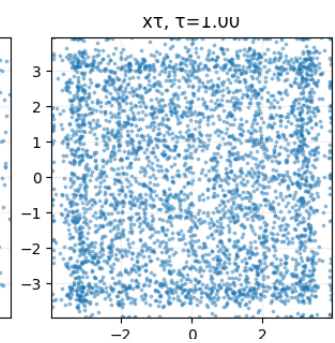
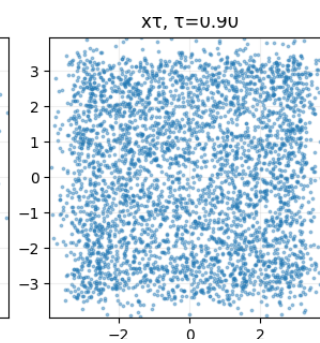
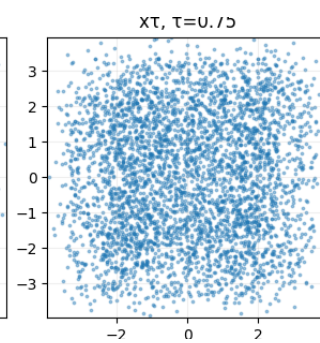
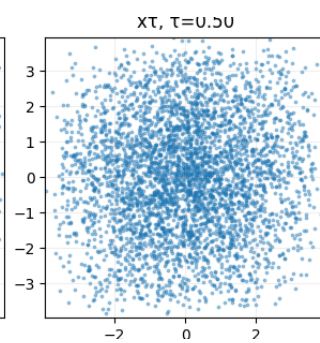
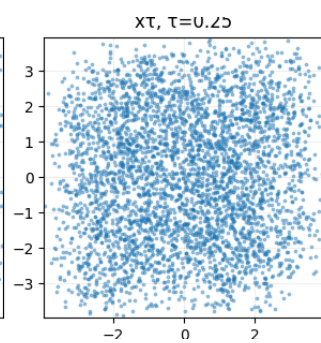
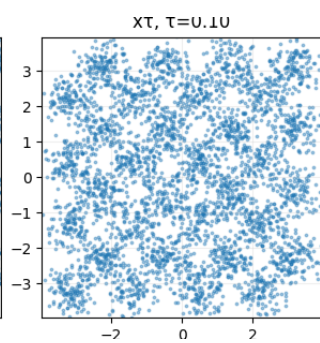
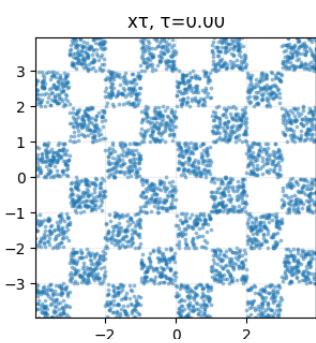


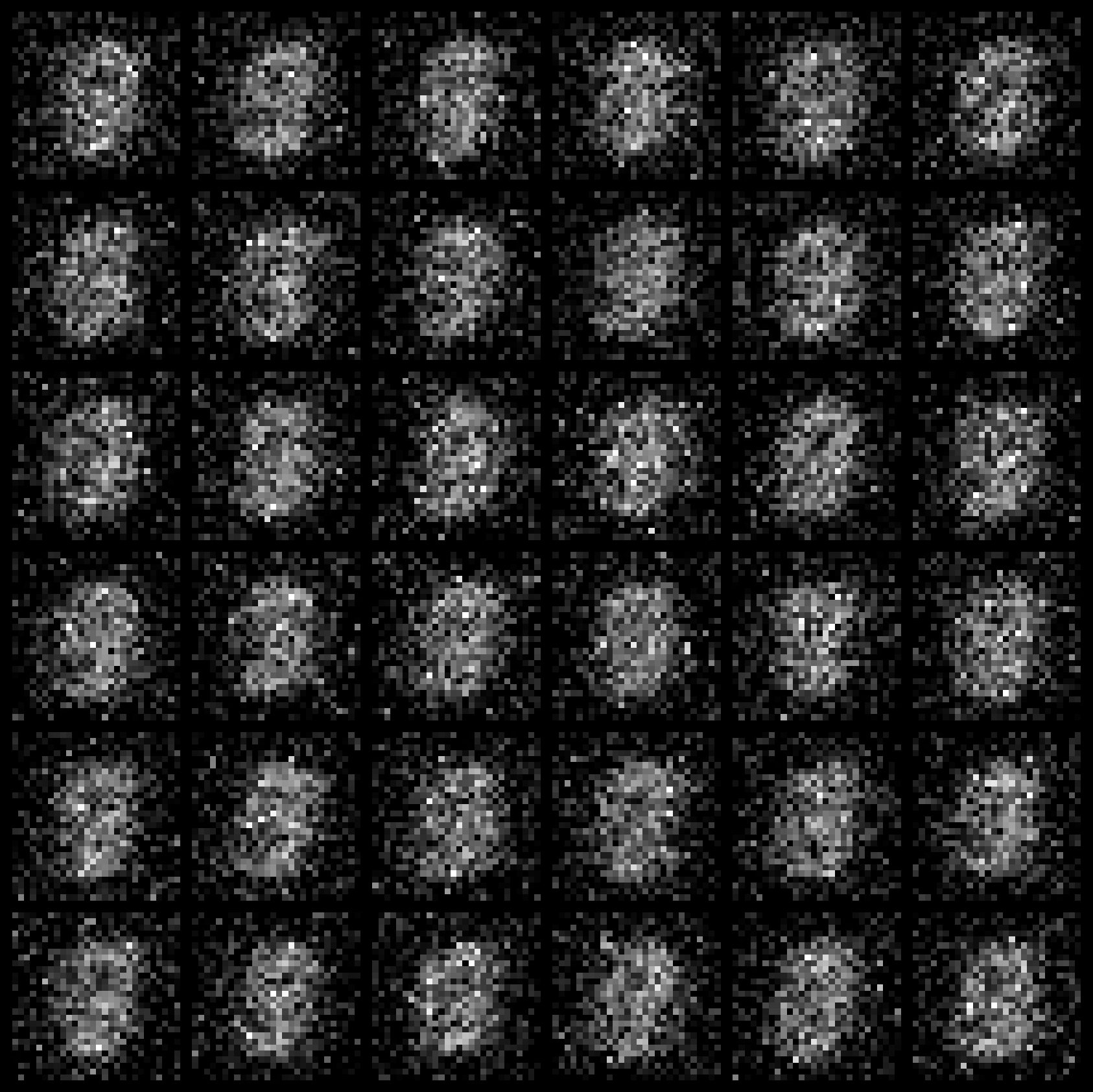




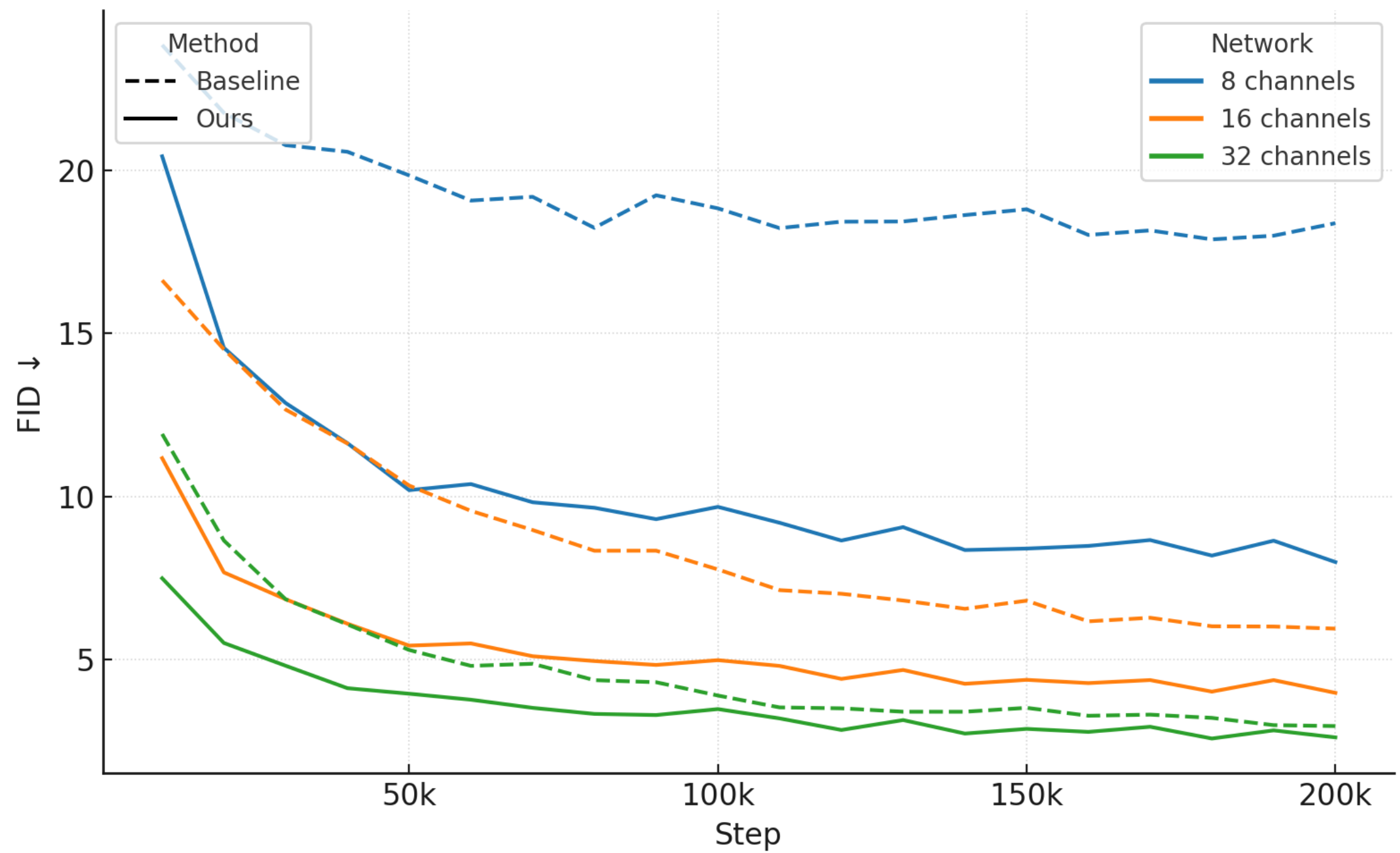


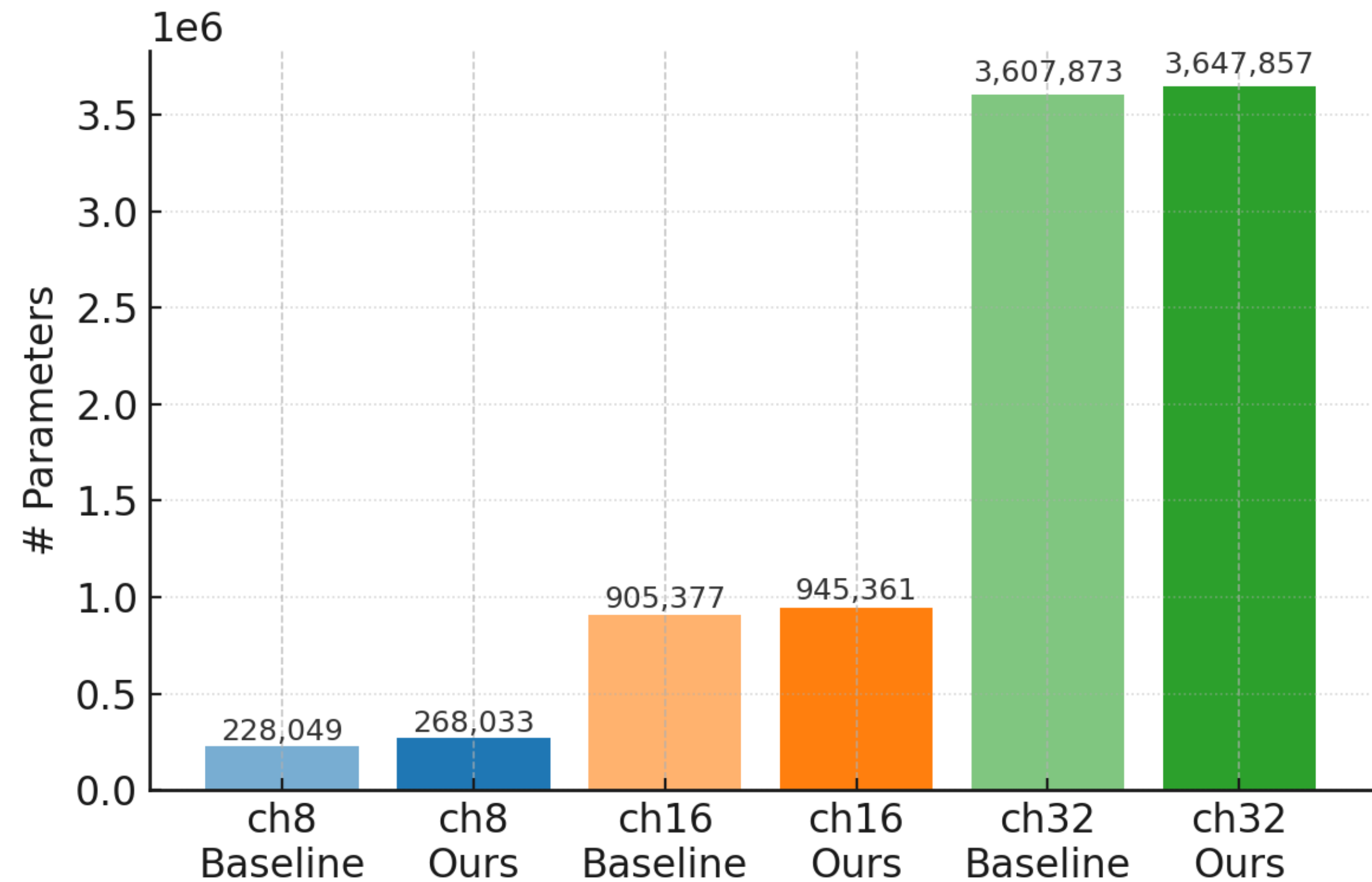




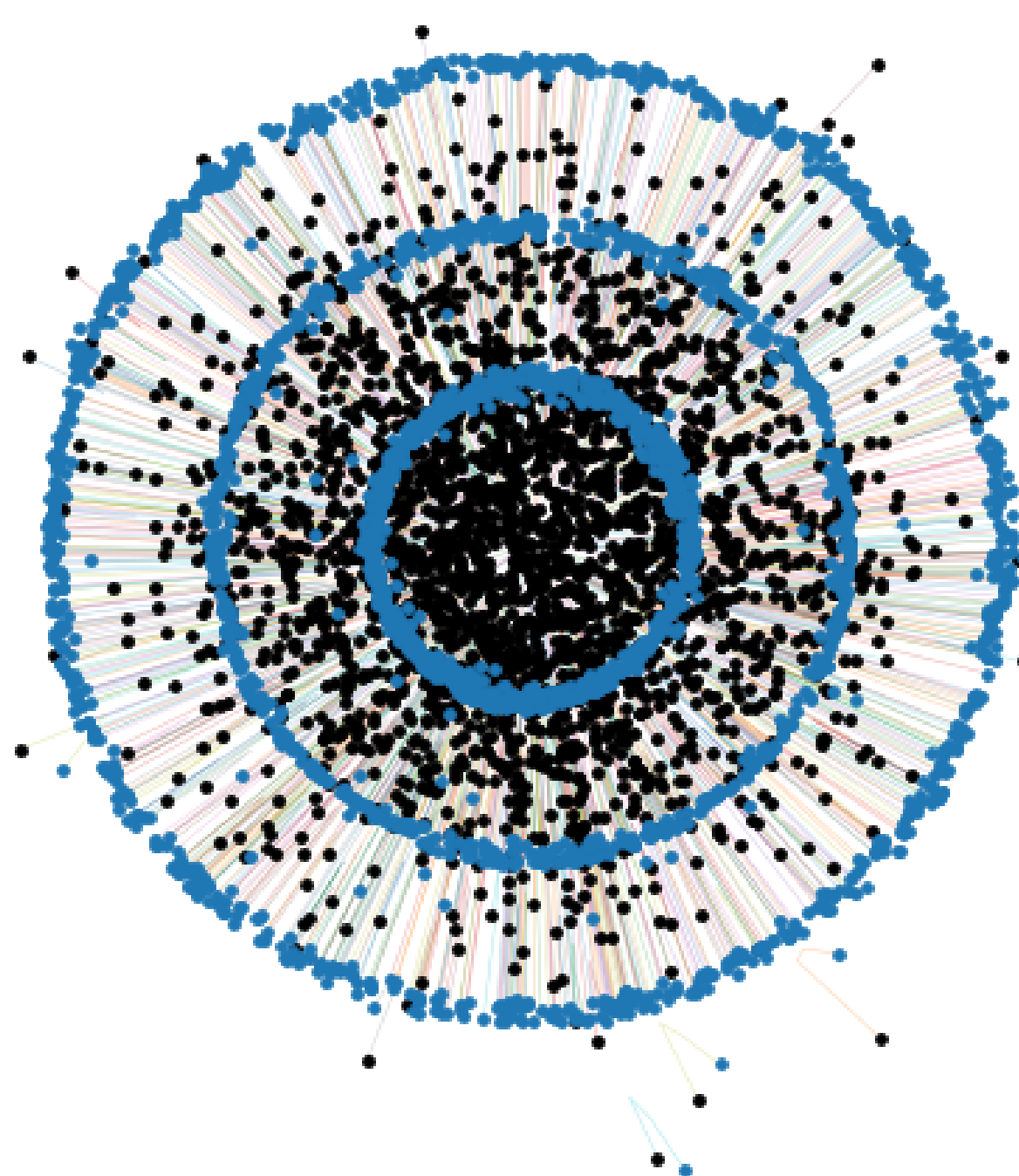


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0	0	9	5	0	7	9	5
8	8	9	1	1	2	5	3
6	2	3	5	9	0	6	0
7	3	6	9	0	9	6	8
8	1	8	2	8	5	6	4
2	0	2	1	3	4	7	8
5	2	8	4	9	1	5	0





Sample paths (start $\rightarrow$ end)



- start ($t=1$)
- end ($t=0$)

Sample paths (start $\rightarrow$ end)

- start ($t=1$)
- end ($t=0$)

